MaaS product objective and service boundary

A proposed product frame for connecting user intent, mobility operations and measurable service outcomes.

01 / PRODUCT OUTCOME & SCOPE

PRODUCT HYPOTHESIS

A person can request, complete and resolve a mobility journey through one coherent service.

Resident / visitor

State destination, timing, access and service needs
Output: consented trip intent with a time window and access constraints.

Mobility operator

Allocate vehicles and manage trip exceptions
Owns matching, exception routing and recovery assignment.

Service provider

Coordinate bookings, capacity and handover
Returns capacity hold, handoff status and observed trip actuals.

City partner

Set service conditions and review coverage
Sets ODD, access and service coverage conditions.

MaaS product layer

C01 identity / API → C02 booking & consent → C03 feasibility / reservation → C04 dispatch & status → C06 support / cases → C07 usage / settlement | C05 energy capability constrains feasible supply

END-TO-END SERVICE LOOP

01	02	03	04	05
Intent	Feasible offer	Confirmed plan	Trip + support	Verified outcome

PRODUCT BOUNDARY

User-facing products

Capture destination, time window, party / access needs and consent; explain eligible offers, conditions, status freshness, support and recovery.

MaaS orchestration

Own booking revision, feasibility, resource bundle, dispatch permit, exception owner and completion evidence. Never infer execution from command submission.

Connected services

Integrate provider, vehicle, site / EMS and city systems through versioned, tenant-scoped contracts; retain source, timestamp, freshness and reconciliation status.

HYPOTHESIS TO TEST

Test the handoff hypothesis by tracing request → provider capacity → vehicle / site readiness → pickup → trip → recovery. For each break, record who detects it, how long it remains unknown, who owns the next action and whether the user promise can still be met. Do not freeze MVP scope until the dominant loss point is observed.

OPEN DECISIONS • Pilot geography / ODD / user segment • Booking and dispatch system of record • Fare, payment and refund owner • Accessibility / language baseline • Provider capacity commitment • Incident and recovery owner • Data retention / consent basis

End-to-end journey across user, operation and platform

One booking identity carries the journey from request through service delivery, exception handling and verified completion.

02 / SERVICE BLUEPRINT



STAGE EXIT EVIDENCE • Plan = valid, consented intent → Request = feasible candidates + binding reasons → Confirm = committed resource / revision → Travel = fresh status + owned exceptions → Complete = observed service actual + passenger outcome

EXCEPTION LOOP

Delay, capacity change or service issue → update ETA / owner → offer a feasible recovery → preserve the original booking and action history.

TRACEABILITY CONTRACT

bookingId + bookingRev + intentId + resource bundle + missionId + observed outcome

Event envelope: tenantId • eventId • bookingRev • source / sourceRev • occurredAt + receivedAt • idempotencyKey • consentRef • quality / freshness. C02 owns booking state; C03 owns bundle feasibility; C04 journals command and observed execution; C06 owns case lifecycle.

ACCEPTANCE INVARIANTS

Confirm only after resource and time-window holds are committed; expire or release holds deterministically. Duplicate intent + idempotency key returns the prior result. Stale availability must return a reason, never a false offer. Lost command acknowledgement → UNKNOWN, query / reconcile before retry. Completion requires vehicle evidence + service actual + passenger outcome. Every open exception has owner, lease / dueAt, next action and escalation.

MVP boundary and product prioritization

Deliver one complete, measurable service slice first; activate broader capabilities as user value and operating readiness are evidenced.

03 / MVP & PRIORITIZATION

Priority rule

Rank by user impact, business value, technical feasibility, risk and learning speed.

Candidate priority is decided with evidence; this document does not claim a validated ranking.

PILOT MVP

One selected service slice

01	Capture intent, needs, consent and idempotent request
02	Return feasible options, binding reasons and expiry
03	Commit booking + resource bundle with revision checks
04	Track state, freshness, support case and recovery owner
05	Reconcile actual trip, service use and passenger outcome

Exit only when normal + recovery paths pass acceptance; operator can resolve exceptions; event and cost records reconcile.

ITERATE

Expand validated value

01	Add segment / service only with distinct evidence
02	Support amend, cancel, no-show and partial failure
03	Reduce avoidable manual handoffs; retain override reason
04	Balance queue, staffing, fleet and recovery workload
05	Tune promise, matching and support from cohort data

Exit when repeat use, reliability, support load and unit economics meet thresholds set from an agreed baseline.

EXTEND PLATFORM

Capabilities by evidence

01	Versioned adapters, tenant / consent boundaries
02	C05 energy envelope in C03 feasibility
03	C04 / C06 guarded assistance and case flow
04	I09 roadside evidence for monitoring only
05	Reusable site config, runbook and conformance tests

Exit when interface owner, data quality, operating authority, safety review, deployment cost and rollback are approved.

MVP CONTROL

MVP entry criteria: selected geography / ODD / channel / segment; named service and incident owners; provider, fleet and site capacity confirmed; legal, privacy, accessibility and partner constraints reviewed; C01–C07 data and authority boundaries agreed. Pilot exit: normal + cancellation + disruption / recovery traces pass; queue and support coverage are staffed; KPI baseline, missing-data behavior and cost ledger reconcile; rollback and customer communication are rehearsed. Optional C05 optimization, remote assistance and I09 roadside sensing stay out until separately gated.
G2 constraint register: ODD = approved operating conditions (road, speed, weather, time, etc.). DDT = real-time operational / tactical functions needed to operate in traffic (including OEDR); record which actor performs them. Define DDT fallback separately by actor, trigger, response and intervention boundary. Also record operating limits (hours / staffing / recovery), infrastructure (node power / energy / footprint) and business limits (fare / demand / transit overlap). For each, record variable, unit / value or TBD, source URL, owner, affected KPI and PL line. Set thresholds only after evidence and ownership are confirmed. Source: SAE J3016_202609, doi:10.4271/J3016_202609.

Interaction concept for user app and operator workspace

Conceptual wireframes translate the service blueprint into testable tasks; they are not final UI or findings from usability tests.

04 / UX FLOW & PROTOTYPE

RESIDENT / VISITOR APP • CONCEPTUAL

01 REQUEST	02 OFFER	03 SERVICE
Plan a journey	**Review options**	**Track and get help**
Origin / destination + pickup point	Mode + provider eligibility	Booking / mission state + age
Arrival / departure window	Pickup / arrival window + expiry	ETA + source freshness
Party size + access / luggage	Access fit + transfer / handoff	Pickup / onboard / arrival proof
Consent + service preference	Fare status + binding conditions	Help, cancel or recovery path
Validate → submit intent	**Compare → accept / revise**	**Confirm outcome / recover**

OPERATOR WORKSPACE • ACTION CONTEXT

QUEUE

Severity / dueAt / age
Passenger exposure + trip stage
Case owner + lease expiry
Unacknowledged / stale signals
Relief capacity + escalation
Filter: site / provider / service

EVIDENCE & STATUS

VehicleId / missionId / bookingRev
CommandId + receipt / execution split
ObservedAt + receivedAt + age + source
Trip state, scene + passenger status
Evidence gaps / map or scene revision
Unknown remains UNKNOWN; replay is read-only

PERMITTED ACTION

Permitted action + reason
Required current evidence
Qualified owner + lease expiry
Expected outcome + timeout
Submit / cancel / escalate
Every action is auditable

USABILITY TASKS TO TEST

Test: submit valid / invalid request; distinguish firm vs conditional offer; amend / cancel; understand stale ETA; recover after lost acknowledgement; request passenger help. Record task success, time, critical error, assistance needed, status comprehension and recovery completion by user segment / accessibility need. Also compare task success and recovery by user segment against request conversion, promise kept and recovery success.

Prototype output: user flow, key states, empty/error cases, accessibility checks and acceptance criteria agreed with Design and Engineering.

STATE MODEL

DRAFT → QUOTED (expiresAt) → PENDING (hold) → CONFIRMED (commit) → IN_SERVICE → COMPLETED (actuals + passenger outcome) / CANCELLED (reason + release). Timeout / lost acknowledgement → UNKNOWN → query and reconcile; never duplicate a side effect. Revision conflict → refresh terms; stale evidence → block offer or action with reason.

Product outcome, KPI tree and instrumentation

Define denominators, event sources and owners before pilot measurement; set target values from a baseline rather than inventing targets.

05 / SUCCESS MEASURES

NORTH STAR

Verified completed journeys

count only when mission evidence + service actual + user outcome reconcile

USER VALUE

Valid-request rate

Valid requests / eligible sessions; segment by need and time

C02 intent + session

Request conversion

Confirmed bookings / valid requests; exclude duplicate retries

C02 intent / booking

Promise kept

On-time pickups / pickups due, including missed pickups

C02 promise + C04 actuals

Trip completion

Completed trips / eligible requests

C04 mission outcome

FULFILLMENT

Feasible matching

Feasible bundles / eligible requests; retain capacity rejects

C03 allocation

Dispatch success

Assigned missions / eligible dispatch requests

C02 requests + C04 assignments

Fleet utilization

Productive hours / available hours; also report scheduled hours

C04 + fleet schedule

Pickup deviation

Observed pickup time − promised window start; median / p95

C02 promise + C04 arrival

SERVICE RELIABILITY

Support resolution

Cases resolved within class target / cases due

C06 case history

Fresh-state rate

Required signals within freshness limit / required signals

C04 / I09 quality

Recovery success

Recovered journeys / disrupted journeys eligible for recovery

C02 + C06 outcomes

Interventions / 100 h

Operator interventions / autonomous hours × 100 h

C04 events + operating hours

ECONOMICS

Revenue / trip

Recognized service revenue / completed trips

C07 contract + ledger

Energy cost / trip

Measured energy cost / completed trips

C05 plan + C07 ledger

Contribution / trip

(Recognized service revenue − attributable energy, vehicle, site, software and operations costs) / completed trips; document cost allocation.

C07 + cost ledger

Energy efficiency

Passenger-km / measured kWh; if occupancy is unavailable, report vehicle-km / kWh separately.

C04 occupancy + C05 meter

KPI → PRODUCT DECISION

Baseline → target range → pilot result → priority change

Review by user segment, service type, site and disruption class; protect against denominator drift and missing-event bias.

TARGETS ARE PENDING

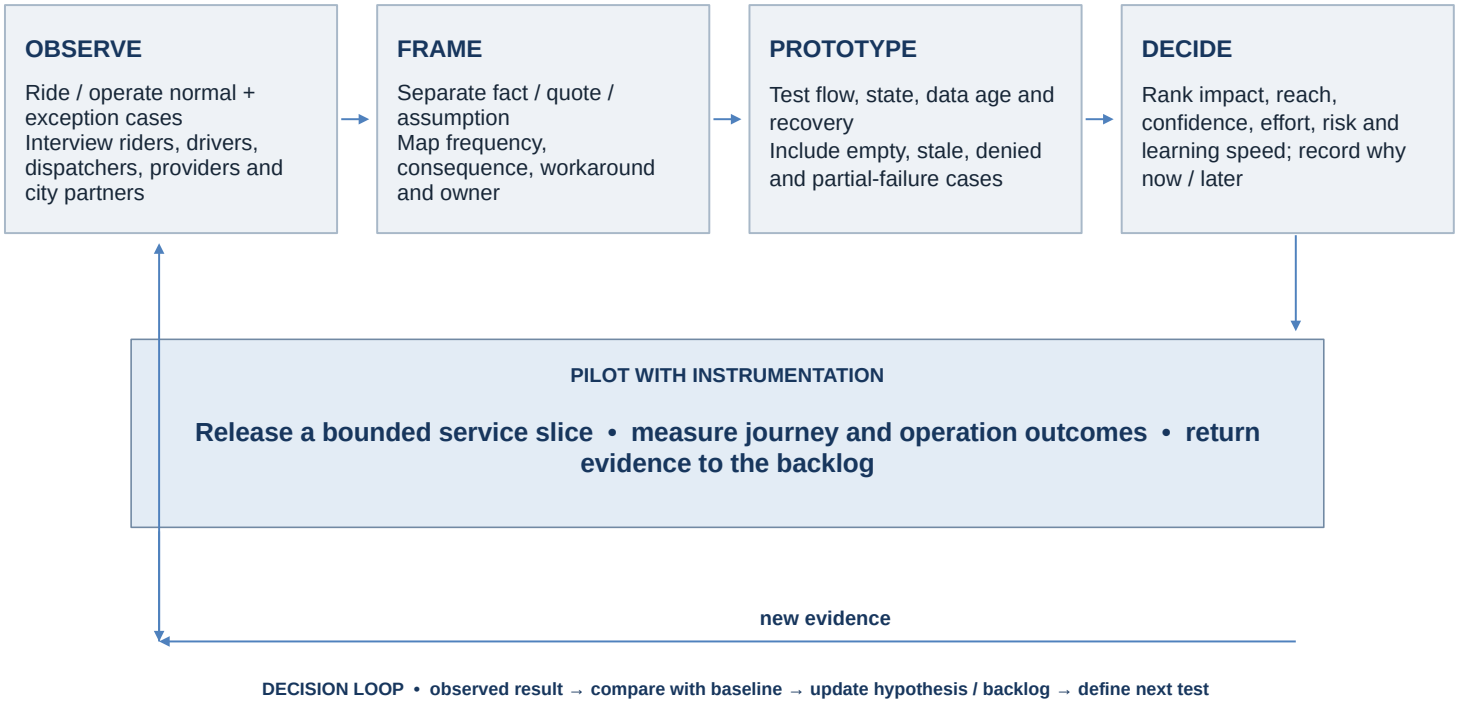
This document contains proposed data contracts, but no pilot baseline or validated target values. Confirm definitions and data access with Operations, Engineering and partner teams.

EVENT / SEGMENT CONTRACT • Use the P2 event envelope and P6 evidence record. Report median / p95 for time metrics; segment by cohort / site / service / disruption; publish exclusions / missingness. Guardrails: command receipt latency p95; ADS dispatch-ready vehicle-hours / scheduled autonomous-service hours; operator interventions / 100 autonomous hours; charging-node occupied / available hours; committed-demand intervals met / scheduled; safety margin by approved limit / scenario with units and direction. Engineering + Operations own technical thresholds; PM owns user / business outcomes. Forecast revenue = forecast completed trips / day × forecast net recognized revenue / completed trip × operating days; actuals use the ledger. Contribution / trip deducts attributable energy, vehicle, site, software and operations costs; document allocation. CAPEX / OPEX separate. Simple node / project payback = first period cumulative operating cash flow ≥ initial CAPEX (undiscounted); show discounted payback separately. IRR solves NPV(r) = Σ CF_t / (1+r)^t = 0 (CF_0 includes investment); use defined cash-flow scope, period, currency and tax basis; compare with the matching hurdle and disclose multiple / no-solution cases. Keep unvalidated inputs TBD with unit, source, owner and calculation link.

Field discovery and AI-assisted product workflow

Use GenAI to accelerate synthesis and drafting while the PM retains evidence, prioritization and release decisions.

06 / DISCOVERY & VALIDATION



GENERATIVE AI SUPPORT • HUMAN REVIEW REQUIRED



SAFE USE RULES

AI may cluster, extract, translate, draft and challenge completeness; it does not create evidence. Use approved tools only; minimize / de-identify personal and restricted data; preserve source links and access controls. PM / domain owner verifies every claim and acceptance rule. AI cannot approve dispatch, widen ODD, set safety limits, authorize a command or close an unresolved case. Measure synthesis time, correction rate, unsupported-claim rate and reviewer acceptance before claiming productivity gain.

Research status: this workflow is proposed; interviews, observation, prototype tests and AI productivity effects remain to be measured.

EVIDENCE RECORD REQUIRED FOR EACH PRODUCT CLAIM

Evidence record: actor / segment • site / service / ODD • occurredAt / source • behavior or verbatim quote • frequency / workaround / consequence • confidence / limitation. Hypothesis: user behavior change + outcome metric + guardrail + disconfirming evidence. Test record: task / context / selection rationale • success condition • completion / time / error / recovery • result and decision owner. AI trace: source IDs • approved tool / model version • prompt purpose • generated claim • reviewer / corrections • accepted or rejected backlog decision. For every KPI or cost input, capture Variable • Value and unit • Source • direct URL • access / extraction date • page or range • formula and usage location.

Roadmap from discovery to repeatable deployment

Progress through evidence gates without fixing dates before site, partner and technical dependencies are known.

07 / RELEASE & SCALE GATES

G1 / DISCOVER

Select segment, site and service context. Observe request-to-outcome in normal and disrupted journeys. Map user goals, operator workarounds, capacity limits, accessibility / consent needs and current alternatives. Baseline only traceable events; state sampling gaps.

EVIDENCE • Journey map and observed friction • KPI definition, denominator, source and missingness • Ranked problems with counter-evidence

GO / HOLD / REVISE

G2 / DEFINE

Choose a bounded end-to-end slice. Define actor needs, system-of-record, service boundary and partner dependency. Prototype normal, cancel, stale-data and recovery paths. Convert evidence into outcome-based backlog and measurable acceptance conditions.

EVIDENCE • Usability results and hypothesis tree • KPI dictionary • Stories with testable acceptance • Open decisions, owner and dependency

GO / HOLD / REVISE

G3 / PILOT READY

Prove C01–C07 contract compatibility and permission boundaries. Validate resource / energy holds, command receipt vs execution, case ownership, staffed support, partner runbook, data quality, privacy review, rollback and user communication. Rehearse failure injection before live service.

EVIDENCE • Pass / fail trace for every critical path • Operating and partner readiness • Known-risk acceptance • Rollback rehearsal and release decision

GO / HOLD / REVISE

G4 / LEARN

Compare cohorts and service classes to baseline. Separate product friction from supply, staffing, policy and data-quality causes. Review guardrails and adverse outcomes. Reprioritize backlog and retest the largest uncertainty. No-change remains a valid decision.

EVIDENCE • KPI delta + confidence / missingness • Incident and case review • Root-cause owner • Experiment result • Continue / revise / stop rationale

GO / HOLD / REVISE

G5 / REPEAT

Package site configuration, partner adapter, service policy, training, monitoring, support and rollback. Confirm local legal / accessibility / operating differences. Forecast marginal onboarding and support effort; preserve configuration without forking the core product.

EVIDENCE • Second-site conformance • Deployment time and cost • Support / reliability envelope • Partner acceptance • Repeatable contribution model and rollout owner

GO / HOLD / REVISE

HOLD / REPLAN RULES

Schedule alone never passes a gate. HOLD for unresolved safety / privacy / legal risk, unknown execution, missing evidence, unstaffed support, partner unreadiness, unreliable data or unreconciled cost.

REVISE with named owner, required evidence, deadline and temporary scope restriction. Return to an earlier gate when new evidence changes the user problem or service boundary. Record rollback trigger and resume authority.

GATE TRANSITION RULE • Gates are evidence checkpoints, not fixed calendar phases. A failed condition returns to the earliest gate whose assumption is invalidated; scope stays held until the named owner closes the evidence gap and the decision is recorded.

PRODUCT DECISION OWNERSHIP

PM: user outcome, scope, priority and KPI decision | Design: interaction and usability | Engineering: architecture and delivery | Operations / partners / city: service readiness and constraints

TECHNICAL EVIDENCE IN THE FOLLOWING SYSTEM DESIGN

Reservation and service contracts: slides 8–11 • Remote operation and roadside interfaces: slides 13–23 • Fault injection and release dependencies: slide 24

Operating capacity, replenishment, recovery and verification conditions follow on slides 25–28. All tests remain proposed.

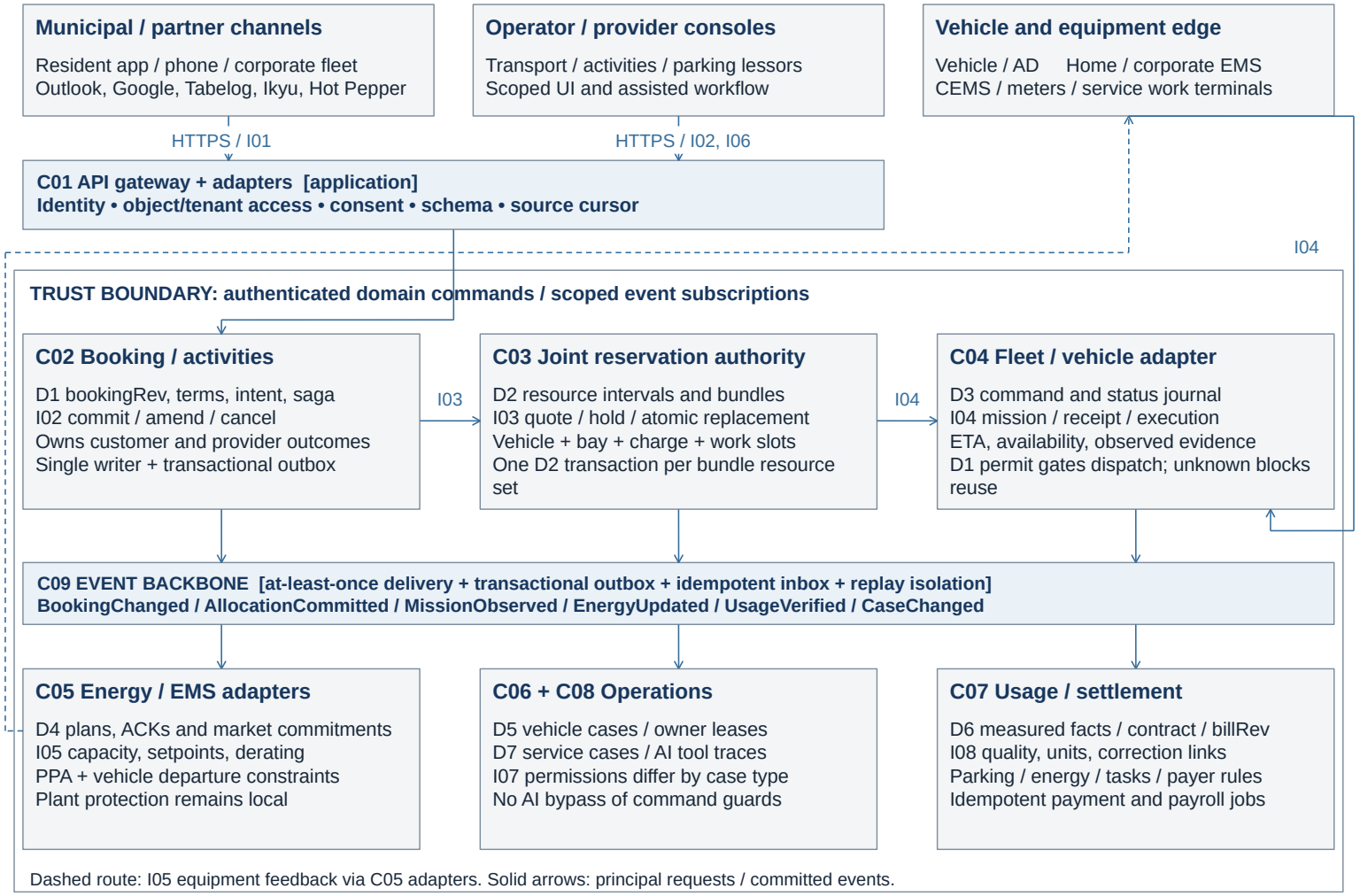
GO / HOLD RECORD

Decision + date • user segment / site • baseline and observed KPI • acceptance trace • unresolved safety / legal / operational risk • scope restriction • accountable owner • next review trigger. GO requires named service owner, trained operation, support coverage, data quality and rollback path. Finance record: CAPEX and OPEX by node and project; node and project payback; IRR; sensitivity range; Variable / Value / Source URL / formula-use link for every assumption.

Container architecture and deployment boundaries

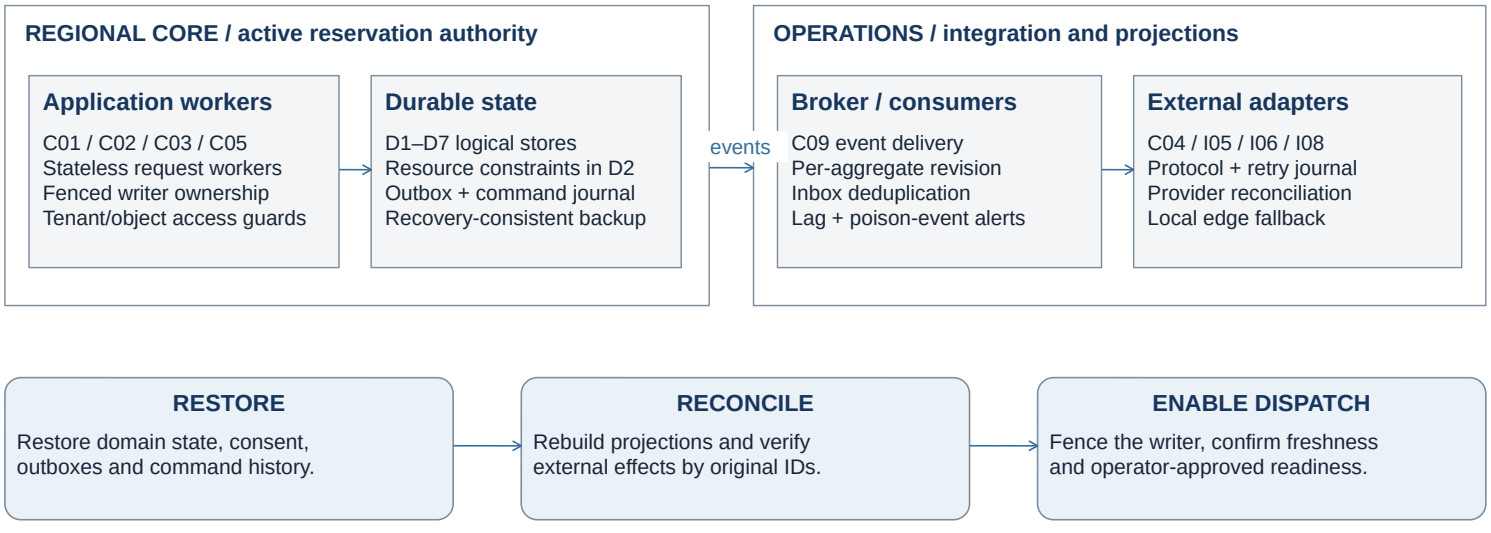
Proposed common PaaS for integrated reservation, dispatch and energy operations. Diagrams define candidate boundaries and contracts. Desk review is complete; implementation and validation remain pending.

1.1 Logical containers, request paths and event distribution



Dashed route: I05 equipment feedback via C05 adapters. Solid arrows: principal requests / committed events.

1.2 Runtime deployment, isolation and recovery



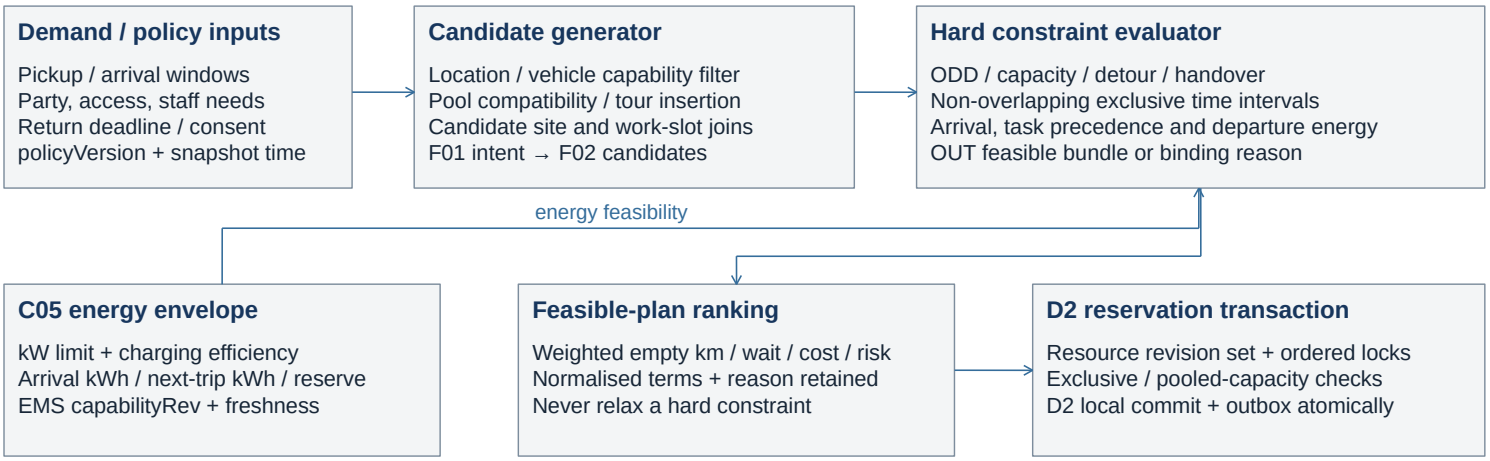
Connector failure may leave local intake available; confirmation still requires all mandatory dependencies. Stale vehicle/site state blocks affected capacity. Replay must use separate projections and idempotent effect handlers.

G2 decisions: hosting zones, failover mechanism, protocols, RPO/RTO, latency and freshness budgets. The primary service integrates booking, dispatch and energy. Equipment/JV/eVTOL options attach through capability interfaces.

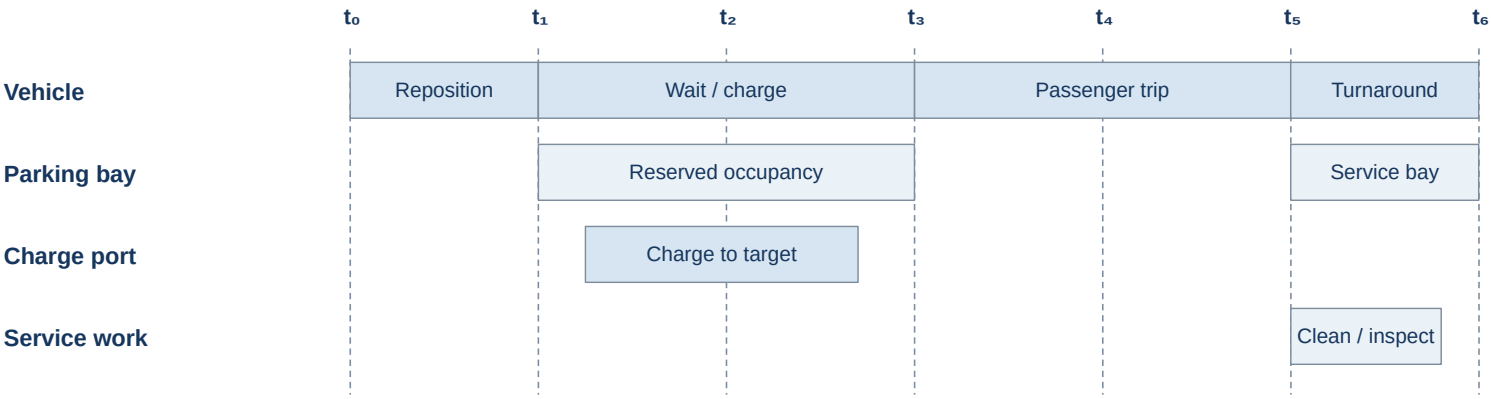
Joint scheduling, resource timing and states

Reservation feasibility includes the complete cycle: pre-position, waiting, passenger service, cleaning or inspection, energy replenishment and the next departure.

2.1 C03 component decomposition and constraint propagation

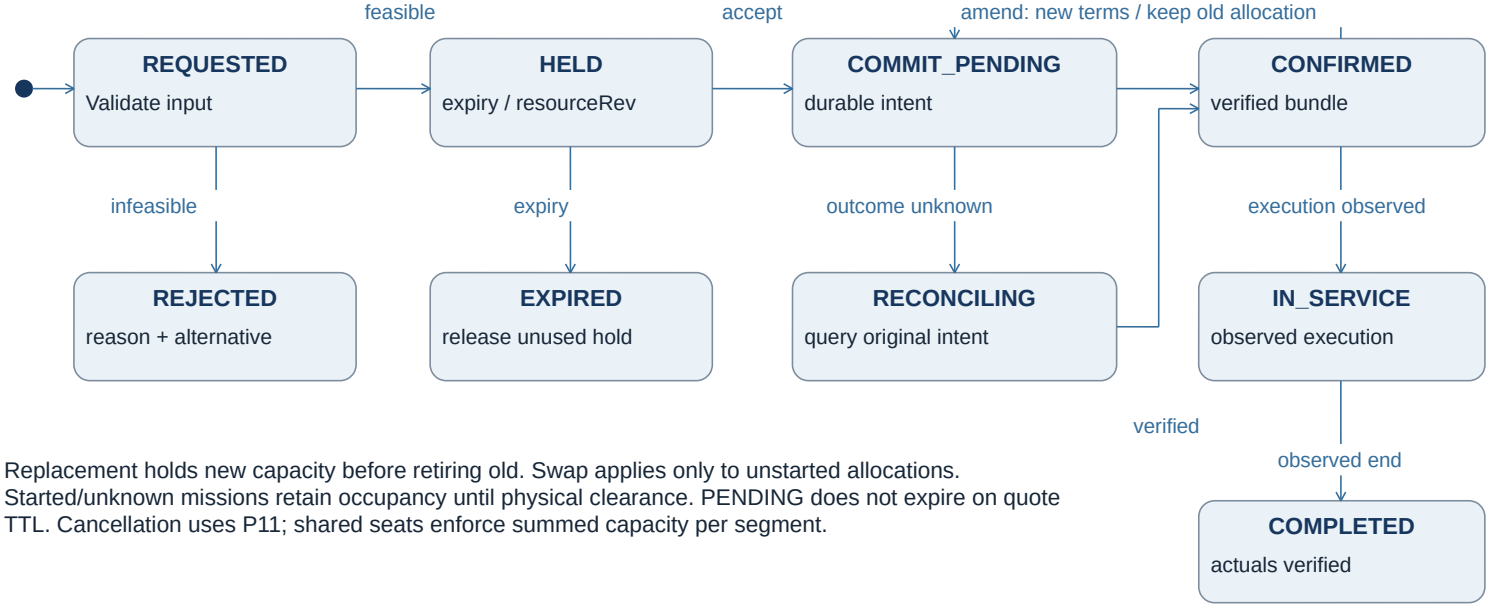


2.2 Coordinated timeline: one vehicle, one bay and one charging session



Departure guard: all required tasks finish before departure; precedence permits only approved overlap. Electrical branch: E_dep = E_arr + Σ(η_ch P_AC Δt_h) − E_aux, with E_trip + E_reserve ≤ E_dep ≤ E_usable,max. kW × hours = kWh.

2.3 Selected reservation transitions and replacement constraints



Replacement holds new capacity before retiring old. Swap applies only to unstarted allocations. Started/unknown missions retain occupancy until physical clearance. PENDING does not expire on quote TTL. Cancellation uses P11; shared seats enforce summed capacity per segment.

Fleet capacity includes empty travel, turnaround and replenishment. Evaluate each zone and time interval against reachable supply. Reserve hub resources separately: bays, connectors, power, work slots and staff. Capacity and coverage checks: P25–P26.

Entity relationships and message contracts

Identifiers, cardinalities, revisions and uniqueness constraints preserve the chain from an accepted journey to resource use and settlement. Example payloads define proposed contracts.

3.1 Domain model with keys and ownership

PlanReference / C01

PK tenantId, planRefId
provider, accountRef, sourceEventId
instanceId, sourceRevision
consentId, purpose, expiresAt
startUTC, endUTC, IANA zone
Unique within tenant:
provider/account/event/instance

0..1 : N

Booking / C02

PK tenantId, bookingId
FK planRefId (optional for app input)
bookingRev, termsRev, state
riderRef, pickupWindow, destination
currentBundleId, intentId
CAS: expectedBookingRev

1 : N

ResourceBundle / C03

PK tenantId, bundleId
FK bookingId, replacesBundleId?
bundleRev, state, expiresAt
policyVersion, inputSnapshotRef
intentId, idempotencyKey, result
Unique: tenant/intent/key

1 : N

1 : N

ActivitySaga / WorkOrder

PK tenantId + sagaId / workOrderId
FK bookingId, bundleId?
providerRef, slotRef, staffRefs[]
conditionsRev, acceptedTermsRev
state, owner, dueAt, evidenceRef
Tasks: cleaning / inspection / handover

ResourceInterval / C03

PK tenantId, intervalId
FK bundleId, resourceId
startUTC, endUTC [half-open]
capacityType, quantity, unit
reservationRev, allocationState
Exclusive: no overlap; pooled: sum ≤ capacity

N : 1

Resource / Capability

PK tenantId, resourceId
kind: vehicle/bay/charger/staff
siteId?, operatorRef, contractRef
capabilityRev, status, observedAt
capacity, constraints, quality
Availability needs current evidence

Bundle references below: 1:N Mission commands • 1:N EnergyPlan revisions • 1:N verified usage and ledger entries

Mission / Command / C04

PK tenantId, commandId
FK bundleId, vehicleId
missionRev, routeRef, validUntil
deliveryState, executionState
statusSeq, observedAt, receivedAt
Same commandId: one immutable payloadHash

EnergyPlan / C05

PK tenantId, planId, planRev
FK bundleId, siteId
capabilityRev, powerIntervals[]
departureMinKWh, expiresAt
acceptedRev, executionEvidence
Plan revision ACK ≠ delivered energy

UsageFact / Ledger / C07

PK tenantId + usageId / ledgerEntryId
FK bundleId, contractRev
meterPointId, interval, qty, unit
quality, sourceRev, payerRef
correctionOf?, billRev, currency
No destructive overwrite of actuals

3.2 Command, result and event examples

I03 CommitBundle / command

```
{
  tenantId: "operator-A",
  intentId: "amend-41",
  bookingId: "bk-18",
  expectedBookingRev: 6,
  newBundleId: "rb-52",
  oldBundleId: "rb-37",
  expectedResourceRevs: [104, 28],
  acceptedTermsRev: 3,
  capabilityRev: "cap-9",
  idempotencyKey: "amend-41"
}
```

Same key + different payload:
reject IDEMPOTENCY_CONFLICT.

I03 Result / guarded outcome

```
{
  intentId: "amend-41",
  state: "COMMITTED",
  bundleId: "rb-52",
  bundleRev: 105,
  intentIdForD1: "amend-41",
  committedAt: "<UTC instant>",
  externalReleaseJobs: ["j-4"]
}
```

Other explicit outcomes:
HELD / REJECTED / PENDING
reasonCode + retryability
Unknown ≠ rejection.

Event envelope / all domains

```
{
  eventId: "evt-902",
  type: "AllocationCommitted",
  schemaVersion: 1,
  tenantId: "operator-A",
  aggregateId: "rb-52",
  aggregateRevision: 105,
  occurredAt: "<UTC instant>",
  receivedAt: "<UTC instant>",
  correlationId: "amend-41"
}
```

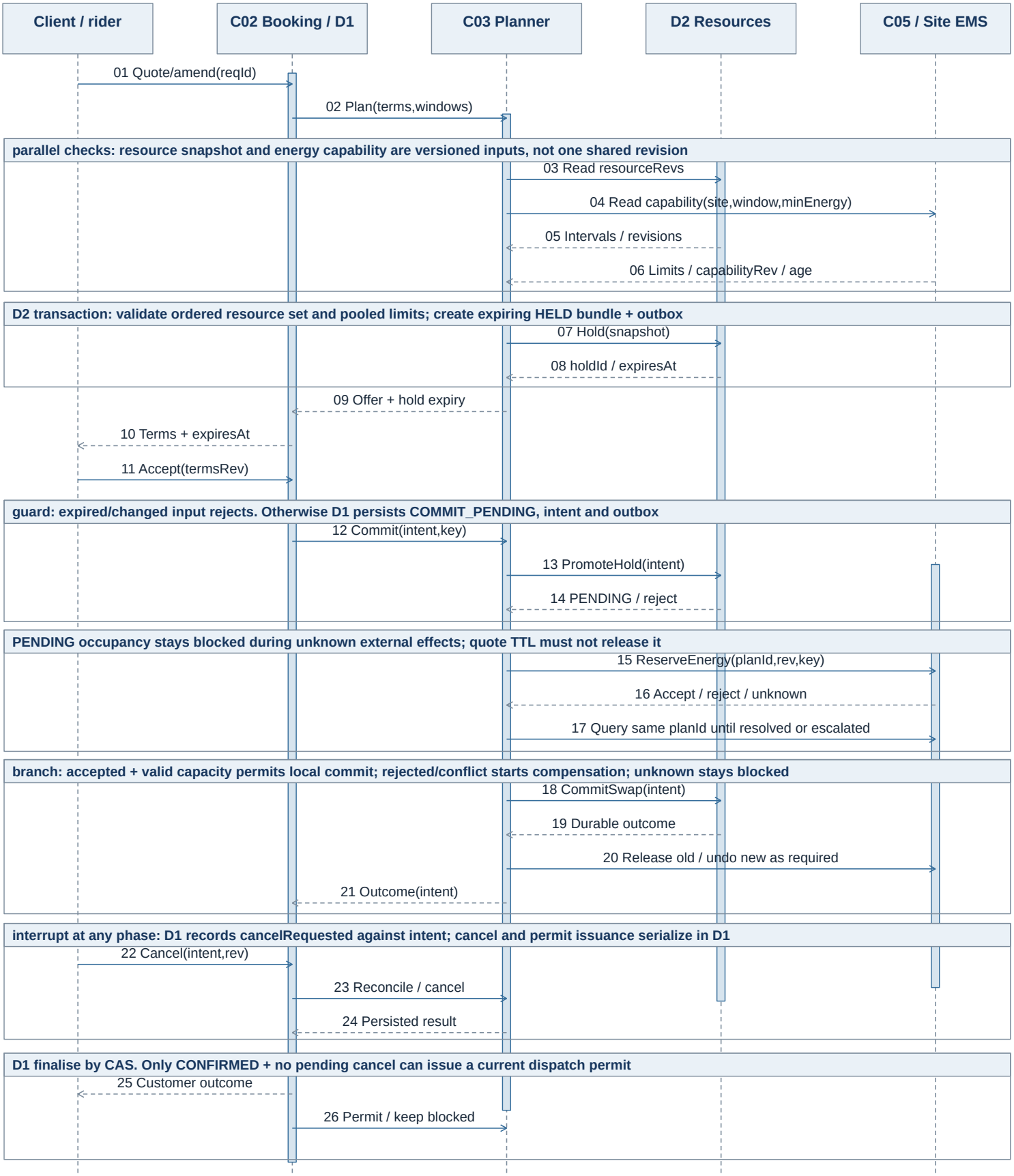
Inbox dedup → revision check
→ apply / fetch gap / quarantine

Keys/FKs include tenant scope; validate references within that scope. expectedResourceRevs maps to the ordered resource set. D1 owns bookingRev. Atomic guarantees cover D2 only. Examples are conceptual payloads, not complete API schemas.

Booking saga, cancellation and dispatch gate

C02 owns booking intent and cancellation. D2 makes only local atomic allocation changes. External EMS effects use a reconciled saga; no distributed atomic commit is assumed.

4.1 Hold promotion, external outcome and authoritative dispatch permit

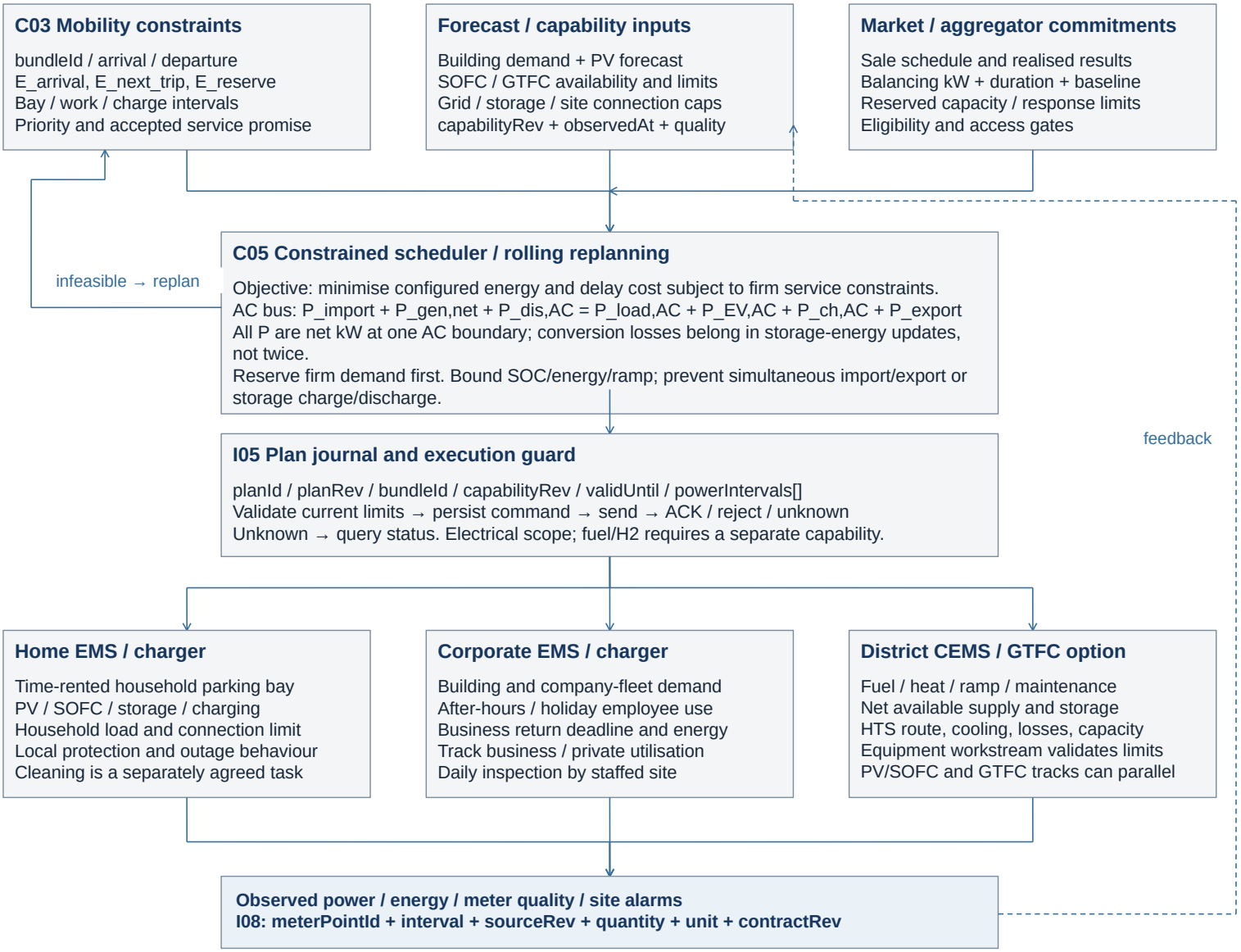


C04 obtains the current D1 permit before releasing a mission. Cancellation after permit issuance becomes cancel/replan of an in-flight mission, not a guarantee of zero movement. Release old physical capacity only after vehicle/EMS clearance. Failed compensation stays owned and reconciled.

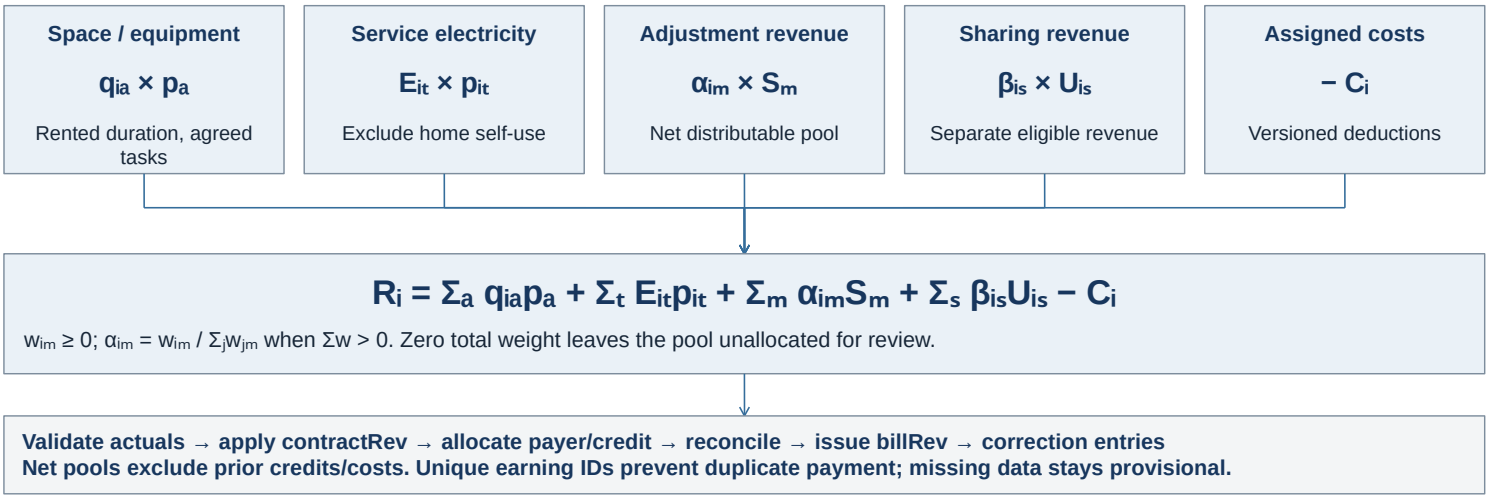
Energy control loops and contribution accounting

The mobility schedule imposes departure-energy constraints. C05 allocates feasible power, reconciles equipment response and sends shortages back to C03. C07 settles measured contributions.

5.1 Hierarchical control and measurement paths



5.2 Settlement dataflow and household contribution formula

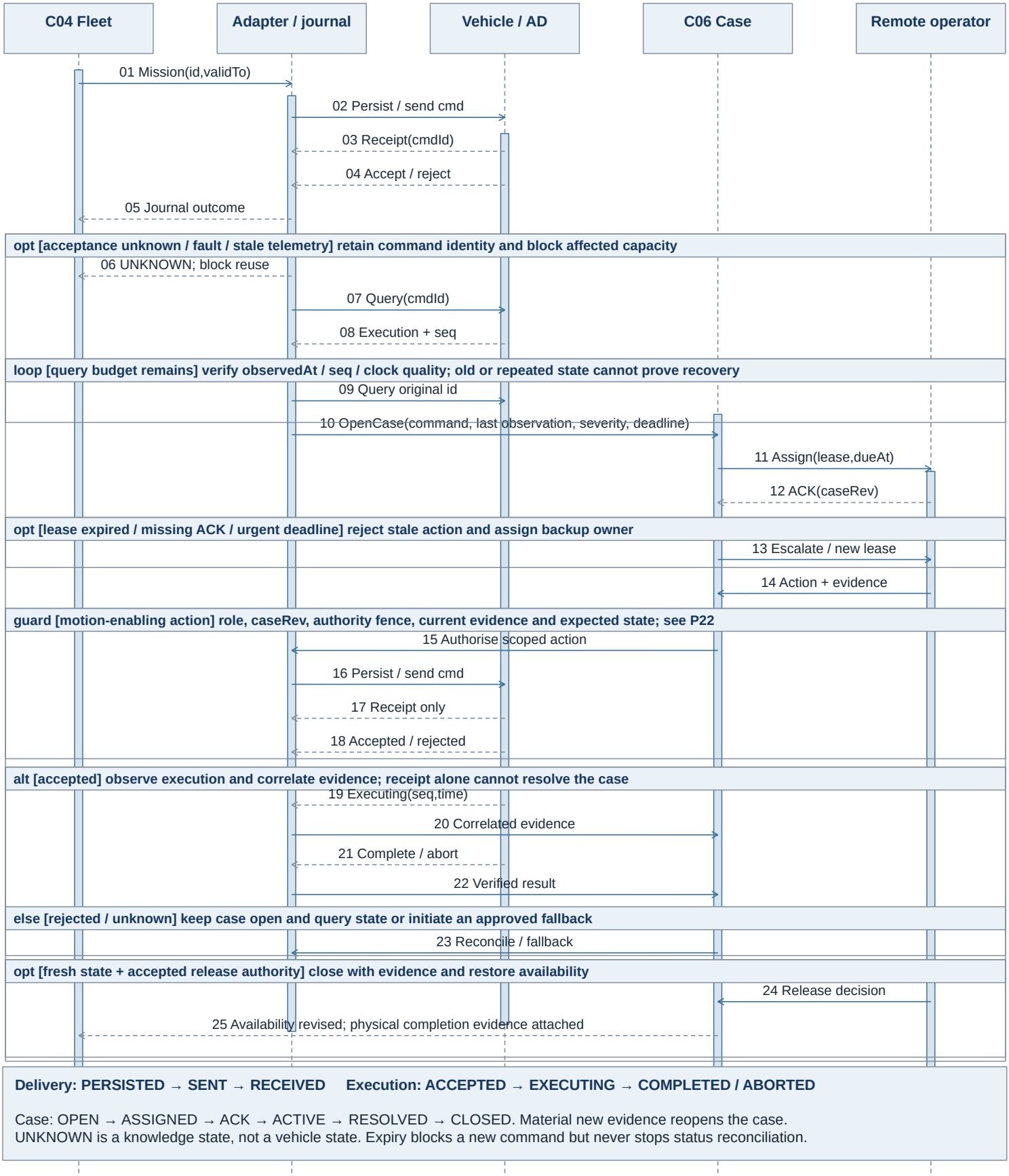


Replenishment queues, site derating and reserve energy constrain each next departure (P26). Settle measured delivery under a versioned contract. Keep net pools, credits and assigned costs non-overlapping. Availability kW and delivered kWh remain distinct.

Remote mission protocol and exception handling

I04 delivery acknowledgment, vehicle execution and I07 case resolution carry different evidence. Permission checks and local vehicle authority constrain every remote action.

6.1 Persistent command, uncertain execution, case ownership and recovery



Delivery: PERSISTED → SENT → RECEIVED **Execution: ACCEPTED → EXECUTING → COMPLETED / ABORTED**

Case: OPEN → ASSIGNED → ACK → ACTIVE → RESOLVED → CLOSED. Material new evidence reopens the case.
UNKNOWN is a knowledge state, not a vehicle state. Expiry blocks a new command but never stops status reconciliation.

Vehicle/ADS owns driving and fallback. Read-only queries and emergency escalation remain available with stale data. Case closure follows P18; motion requests use P22 action-specific guards. Rejected/unknown effects stay reconciled.

Remote architecture and control authority

RM-01 / RM-02. Level 4 ADS retains DDT and fallback authority. Remote monitoring, contextual assistance and passenger support use separate permissions and evidence paths.

7.1 Onboard functions, remote channels and command boundary

VEHICLE TRUST ZONE / device identity, boot epoch and capability revision

ADS / local safety authority

OEDR: detect and respond to road events
DDT: tactical / operational driving
Perception, localisation, motion control
DDT fallback initiates local response
MRM seeks MRC with achieved-state evidence. Zero speed alone is not MRC.

Edge adapter / durable journal

Vehicle state, ODD and diagnostic feed
Command identity, expiry and dedup
Receipt / acceptance / execution split
Safety checks remain onboard
Prioritised uplink + event buffer
No cloud override of ADS constraints

Media / passenger interface

Exterior views and calibration IDs
Scene / frame capture timestamps
Optional approved cabin channels
Passenger SOS / accessible voice
Independent media-health evidence
No stale frame rendered as live

Telemetry / alarm channel

Reliable events + heartbeat + seq
Gaps, clock quality, age and validity
A missing signal is explicit uncertainty.

Command / result channel

Authenticated, scoped, expiring action
Current lease + capability + scene
Onboard accept/reject + observed result

Evidence / media channel

Adaptive video / keyframe evidence
Separate channel health and age
Replay mode disables live actions.

C04 vehicle adapter / I04

Validate schema, identity, unit and seq
Project state with observation quality
Journal command effects and unknowns
Preserve raw events and evidence gaps
Query original IDs after lost responses
Never default missing state to healthy

C06 cases / I07 policy

Correlate vehicle + I09 roadside evidence
Qualified seat + single action lease
Bounded RA advice / escalation
Preserve source quality and conflicts
Fresh vehicle evidence gates release
Service AI cannot authorise motion

Operator / PaaS consumers

Focus scene, history and evidence age
C03 holds affected vehicle capacity
C02 revises journey / passenger offer
C05 revises feasible energy / site slots
No automatic re-entry on reconnect
No release merely because case closed

Roadside sensors / RSU / edge → C10 I09 validation → C06 monitoring evidence [details P15–P16]

7.2 Operational roles and failure containment

Remote assistance and remote driving

RA supplies contextual advice that the ADS can reject after re-evaluating the scene and ODD. Real-time remote DDT is outside this baseline. Adding remote driving requires a distinct operating envelope, control arbitration, communications safety analysis and operator / approval evidence.

Operational and legal responsibility

Map system roles to the permitted operation, designated supervisor and field-response organisation. Passenger support cannot override door interlocks or authorise motion. Vehicle-specific link-loss and fallback behaviour remains local and does not depend on a remote reply.

Active authority

Durable cases / command journal
Replicated evidence + backlog
Vehicle fence ACK gates new actions

Standby promotion

Acquire higher authority epoch
Require vehicle fence ACK (P20)
Reconcile prior effects before action

Degraded service

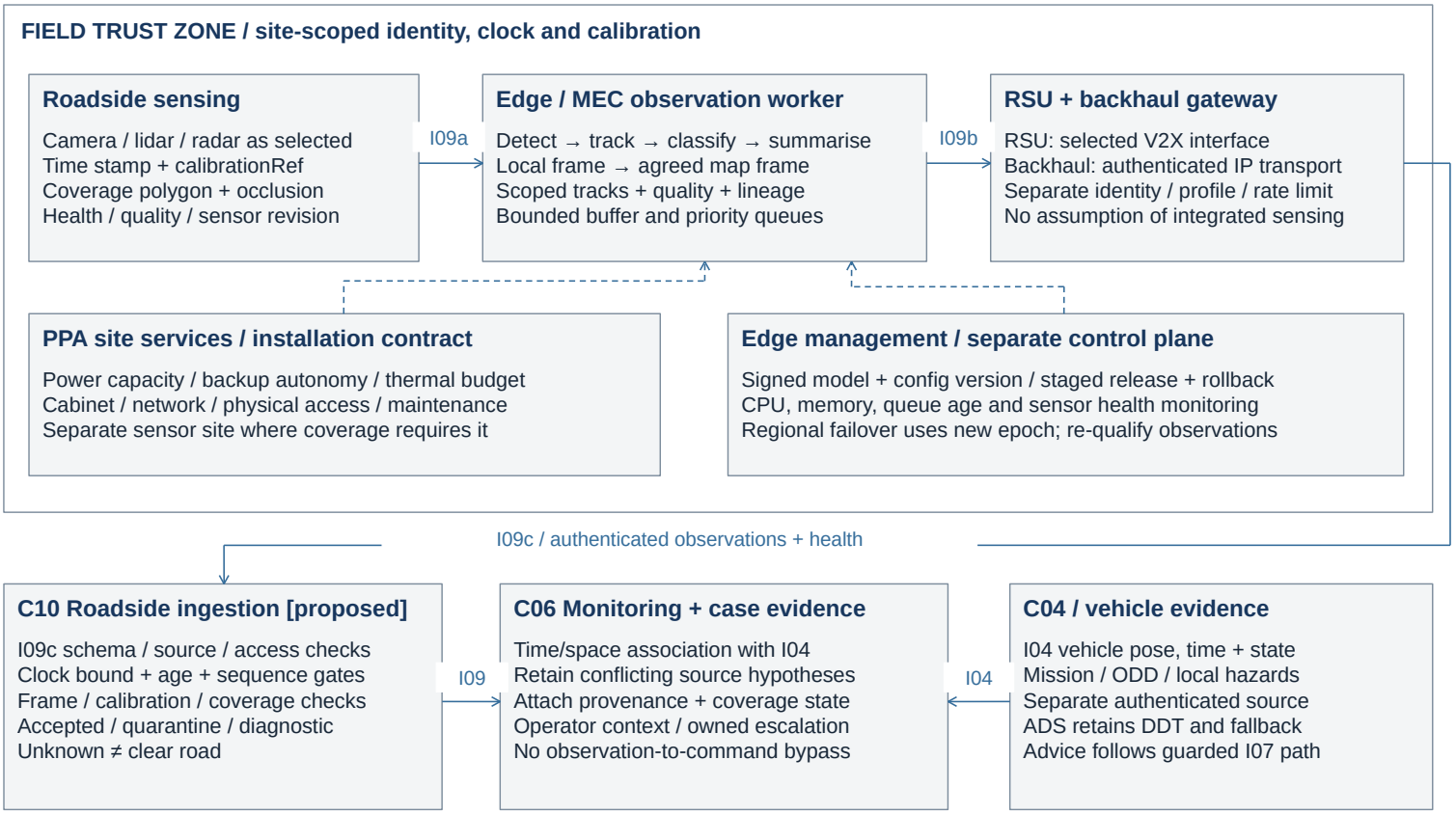
Hold affected assignments
Preserve priority safety traffic
Escalate without hiding uncertainty

Acceptance: independently partition telemetry, media, primary centre and workstation. Old-epoch requests reject after the vehicle installs its fence. Test durable deduplication and unknown outcomes. Reconnect and case closure require the release predicate on P18 before availability can change.

Roadside sensing and edge computing architecture

RM-13. Roadside observations supplement vehicle evidence. Sensor, compute and communication functions have separate ownership. Proposed I09 feeds monitoring and operational review, not direct actuation.

8.1 Deployment topology and data / management plane separation



8.2 Proposed observation contract and association keys

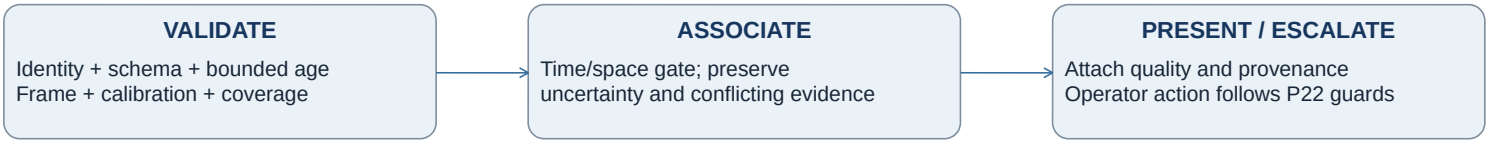
ObservationEnvelope / I09c

tenantId, siteId, sourceId, sensorIds[], bootEpoch, seq
observedAt, emittedAt, clockErrorBound_s, expiresAt
trackKey = sourceId / bootEpoch / localTrackId
frameId, mapRev, calibrationRev, coverageRef
position_m, velocity_mps, uncertainty / covariance
class + confidence, visibility, modelRev, evidenceRef
Signature / authenticated transport + payloadHash

CoverageHealth + association evidence

coverageId / polygon / exclusions / sampledAt
Sensor availability, clock validity, calibration status
Occlusion / weather / queueAge / processing status
Vehicle association: spatial-temporal gate + uncertainty
Candidate link has confidence, method and expiry
No global identity assumed from local track IDs
Multiple reports retain source lineage to avoid double count

8.3 Evidence eligibility, overload and storage boundaries



Invalid / stale / overloaded

Quarantine observations, publish coverage degradation.
Reserve queue capacity for health / critical events.
Drop policy is explicit; historical replay cannot act live.

Evidence / privacy / regional placement

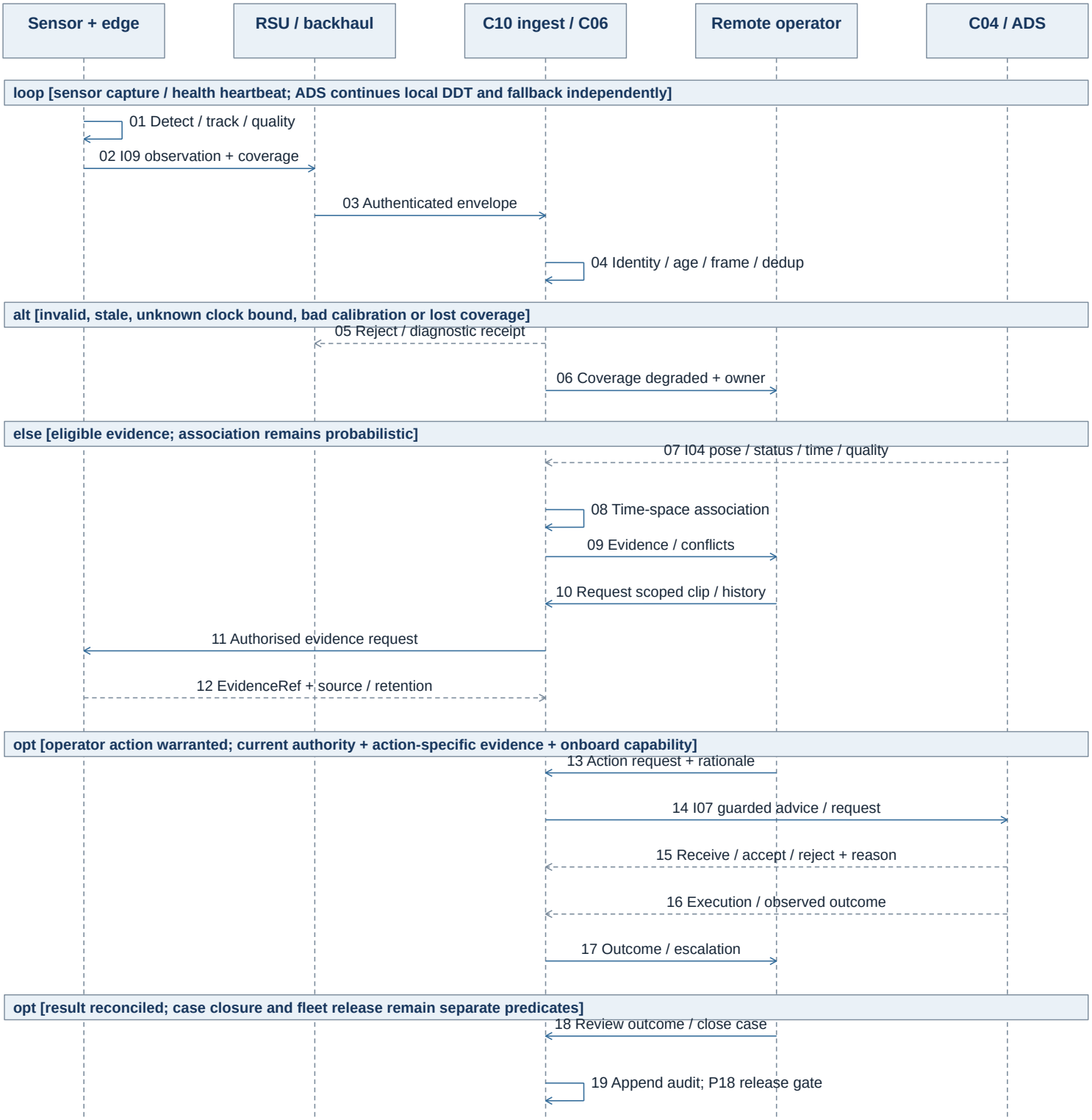
Raw clips by purpose + access + retention contract.
Edge summaries first; authorised evidence on request.
Regional workers scale independently of ADS control.

Thresholds for age, association, coverage and queue size are agreed per use case. CPM mapping and radio profile require a separately verified adapter. Proposed monitoring path does not claim collision avoidance or perception completeness.

Roadside observation and remote monitoring sequence

RM-14. A roadside event becomes scoped monitoring evidence only after validation. The sequence retains independent ADS behaviour, conflicting evidence and unknown outcomes.

9.1 Observation, evidence correlation and guarded response



9.2 Fault branches and planned verification

LINK / EDGE LOSS
Heartbeat expiry → coverage UNKNOWN; retain source time.
No stale frame displayed as current. Reconnect uses new epoch, resync + health checks; replay is historical only.
Test disconnect, reboot, reordering, duplicate delivery.

CONFLICT / OVERLOAD
Keep both source claims; no silent override of ADS state.
Overload reports queue age and dropped observations.
Test occlusion, weather, false association and saturation.
Measure quality, end-to-end age and operator workload.

Telemetry coverage and alarm decision logic

RM-03 / RM-04. Capability-specific signals feed versioned detection and triage. Alarm priorities are operational response classes, not ASIL classifications. Every action uses current, valid evidence.

10.1 Signal families, decision context and missing-data behaviour

ADS / ODD	adsMode, engagementState, oddVerdict/reason, oddProfileRev Route / geofence / weather / illumination constraints as supported.	ODD or mode inconsistency opens a case. Cloud context never widens ODD.
Motion / localisation	pose + coordinateFrame, speed, heading, localisationHealth Covariance and integrity bounds remain distinct supplier quantities.	Detect unexpected stop / deviation. Uncertain pose cannot prove location.
Perception / planning	sensorHealth, occlusionFlags, plannerStatus, blockageReason assistanceRequest + sceneRev + mapRev + frame capture time	Review ambiguity in context. Generic “AI confidence” cannot permit motion.
Chassis / fallback	brake / steering / propulsion health, emergencyStopStatus fallbackReason, mrmState, mrcAchieved, standstill / secureState	Separate manoeuvre progress from achieved MRC. Failed fallback escalates.
Passenger / mission	missionId/rev, pickup/drop-off state, door/ramp/restraint status SOS, occupancy quality, accessibility and handover exceptions	Local interlocks prevail. Unknown occupancy requires supported confirmation.
Energy / link	SOC, usableEnergy, HV/thermal fault, departure reserve margin heartbeat, RTT, loss, jitter, mediaAge, seqGap, clockQuality	Energy shortage revises feasibility. Link loss marks remote state UNKNOWN.

10.2 Evidence validation, alarm lifecycle and response classes

Validate / detect

Identity, schema, unit, epoch/seq
Age / source quality / clock bound
Predicate + persistence + hysteresis

→

Correlate / assign

Vehicle / cause / incident window
Severity + exposure + dueAt
Qualified seat + ACK + lease

→

Act / verify / clear

Guard action and observe outcome
Stable clear condition + release
Reopen on contradictory evidence

P0	Collision / fire / SOS / failed fallback. Immediate escalation, no debounce.
P1	ODD exit / MRM / critical fault / unknown active state. Hold capacity, urgent response.
P2	Assistance / obstruction / boarding failure. Assess exposure and escalate if worsening.
P3	Noncritical service or diagnostic issue. Schedule work under explicit restrictions.

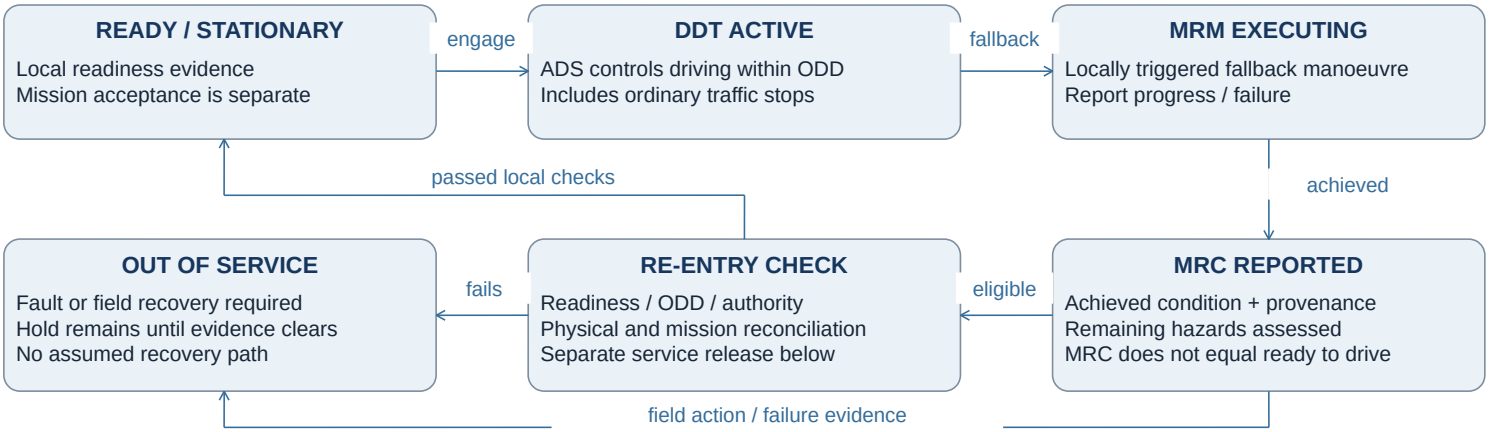
Vehicle I04 / roadside I09 remain separate source envelopes. I09 adds coverageRef, frame/map/calibrationRev, clock bound and source lineage. Preserve original observation time; repeated or replayed samples never refresh it.

Clear only after valid samples span an agreed stable interval. ACK does not clear health. Common-cause grouping retains vehicle exposure and deadlines. Snooze needs reason/expiry and cannot suppress P0. Missing/invalid data raise their own event.

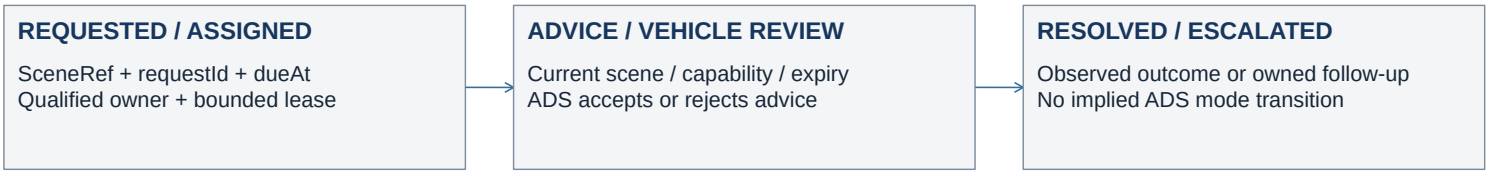
Orthogonal states and guarded return to service

RM-05. ADS operation, assistance workflow, remote knowledge and fleet eligibility are separate dimensions. An ADS can continue its approved local behaviour while a remote assistance request is pending.

11.1 Vehicle-reported operating state: selected transitions



11.2 Assistance lifecycle runs independently of ADS operation



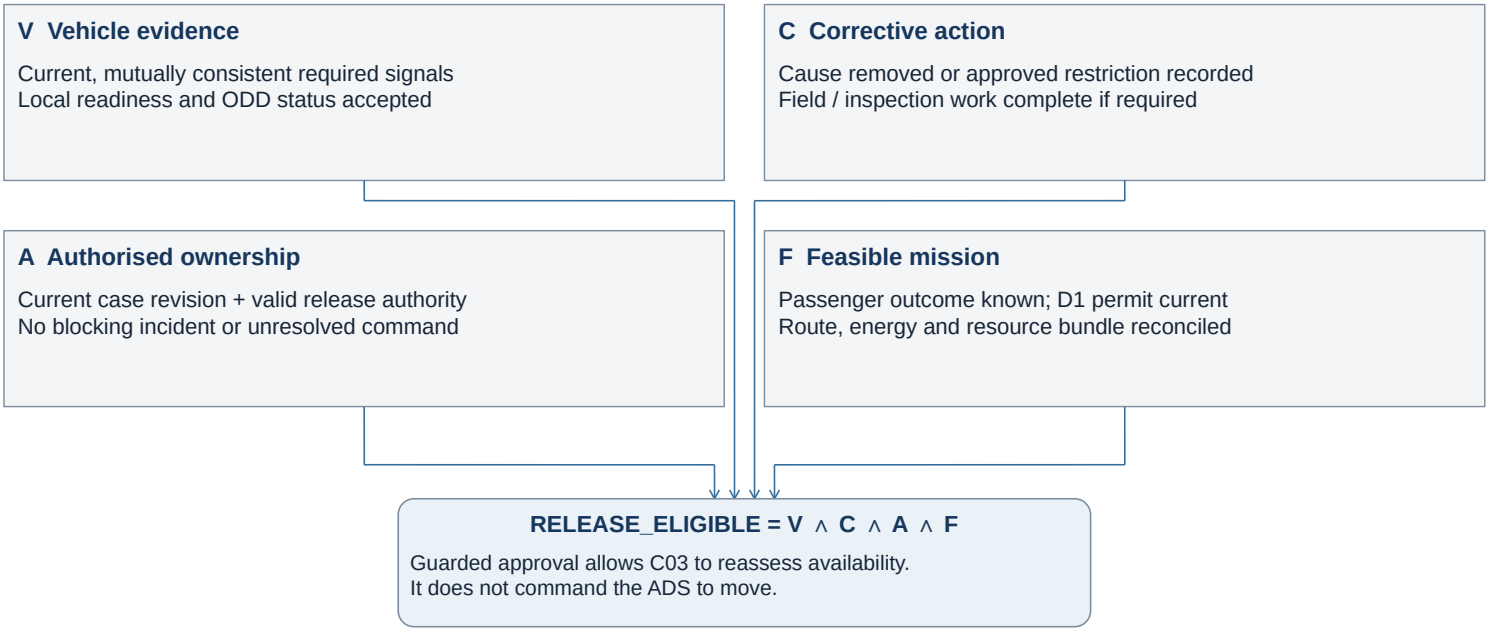
Remote knowledge

CURRENT / DEGRADED / UNKNOWN is per required signal. Fresh data may still report a fault. Unknown physical execution blocks conflicting reuse.

Fleet eligibility / case state

AVAILABLE / RESTRICTED / HELD differs from OPEN / ACTIVE / CLOSED. Closing a case never changes ADS mode or automatically clears an operational hold.

11.3 All release predicates must hold; none substitutes for another

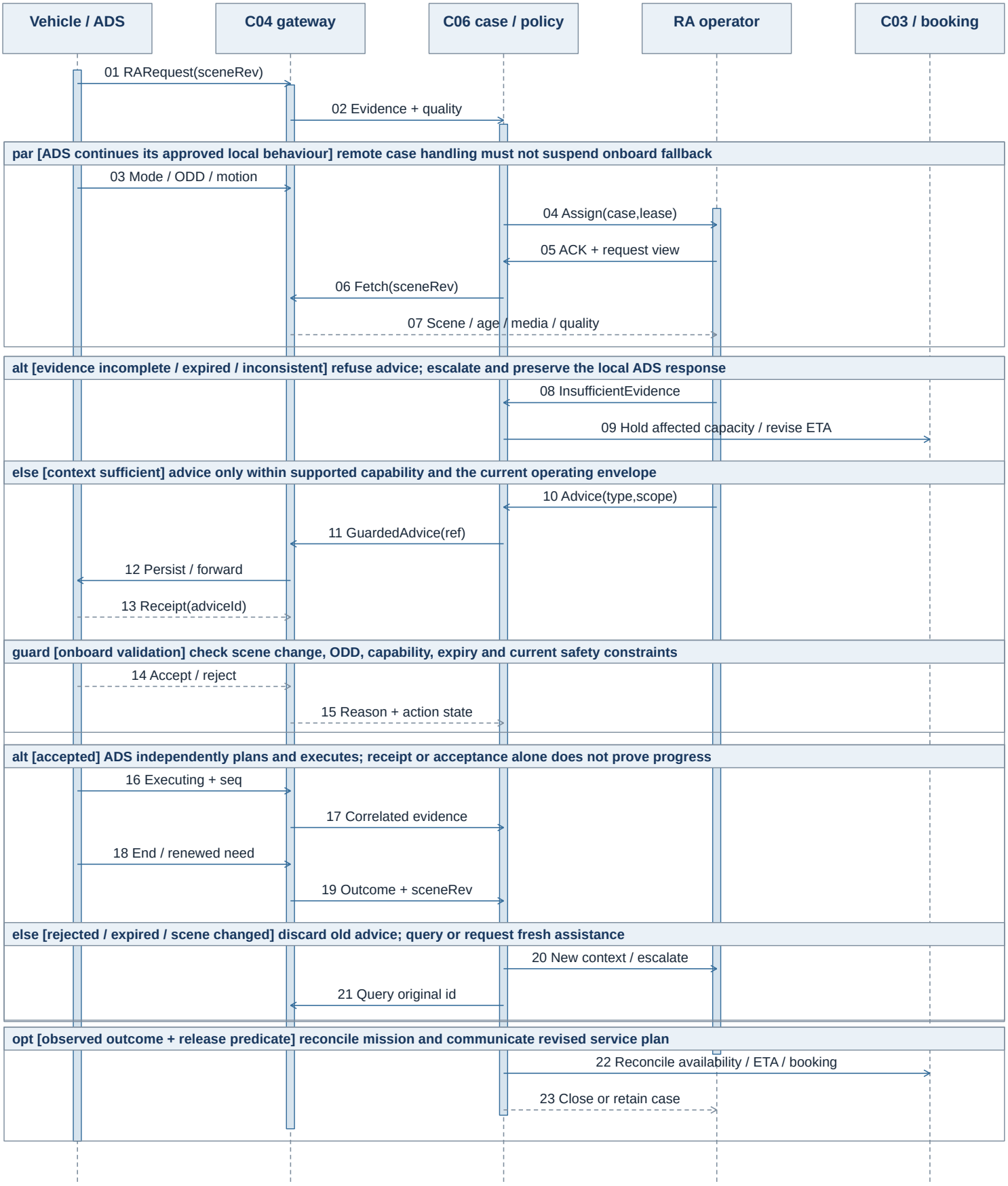


Test cross-products, not only one state chain: DDT_ACTIVE + RA_PENDING; MRC + stale evidence; CLOSED case + critical fault; fresh telemetry + unresolved command. A reported zero speed never suffices for MRC or restart.

Remote assistance: contextual advice sequence

RM-06. Example: the ADS encounters an ambiguous obstruction. A bounded advice transaction carries a scene reference and expiry. The ADS retains trajectory generation, collision checks and motion control.

RM-06A Request, evidence, advice validation and observed outcome

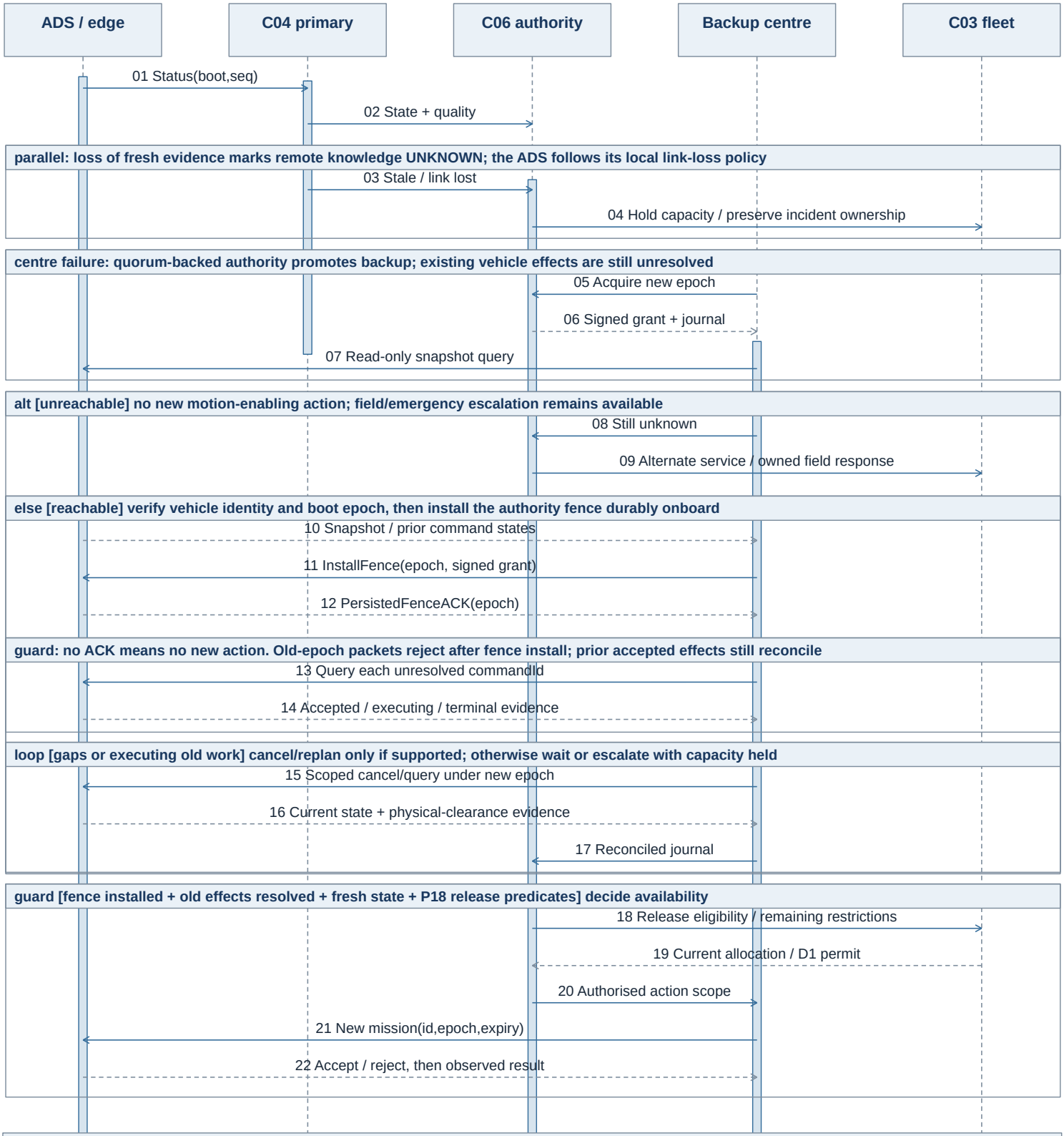


RM-06 invariant: an advice response cannot widen ODD, disable collision avoidance or create a direct actuator path. Timeout/rejection requires reassessment; fresh autonomous resolution can close the case. Executing effects retain their identity.

Link loss, authority fencing and reconciliation

RM-07. Centre authority and vehicle execution require separate evidence. New cloud ownership does not instantly revoke an unreachable vehicle's old authority. Unreachable vehicles remain held.

13.1 Centre change with per-vehicle fence installation



Fence contract: vehicle durably stores its accepted authority epoch and validates signed grants, expiry and command identity. Reboot without recoverable authority/journal evidence blocks new action until resynchronised.
Centre ownership loss blocks new dispatch, but cannot erase an already accepted physical effect. No automatic resend with a new ID. An unsupported fence/dedup capability excludes automated centre handover for that vehicle.

Operator workspace and incident response

RM-08 / RM-09. A focused workspace links fleet triage, ADS evidence, passenger contact and field recovery. Each workstream retains a qualified owner and its own authority.

14.1 HMI wireframe: evidence, authority and operator handover

CENTRE ACTIVE / epoch 42 **Operator O-17 / qualified RA** **Case C-204 / lease current**

Fleet / incident queue

Severity / vehicle / dueAt
Passenger exposure / owner
Oldest unacknowledged case
Common-cause incident link
Seat qualification and health
Escalation always accessible
Colour is not the only cue.

Scene / media / synchronised history

Exterior views / pose / route / scene objects
Map and calibration revision / ADS intended path
LIVE, DELAYED or REPLAY with capture age
Missing view explicit / source quality visible
Align observedAt and receivedAt / mark gaps
Replay disables action. Annotation keeps author, time, evidence reference and case revision.

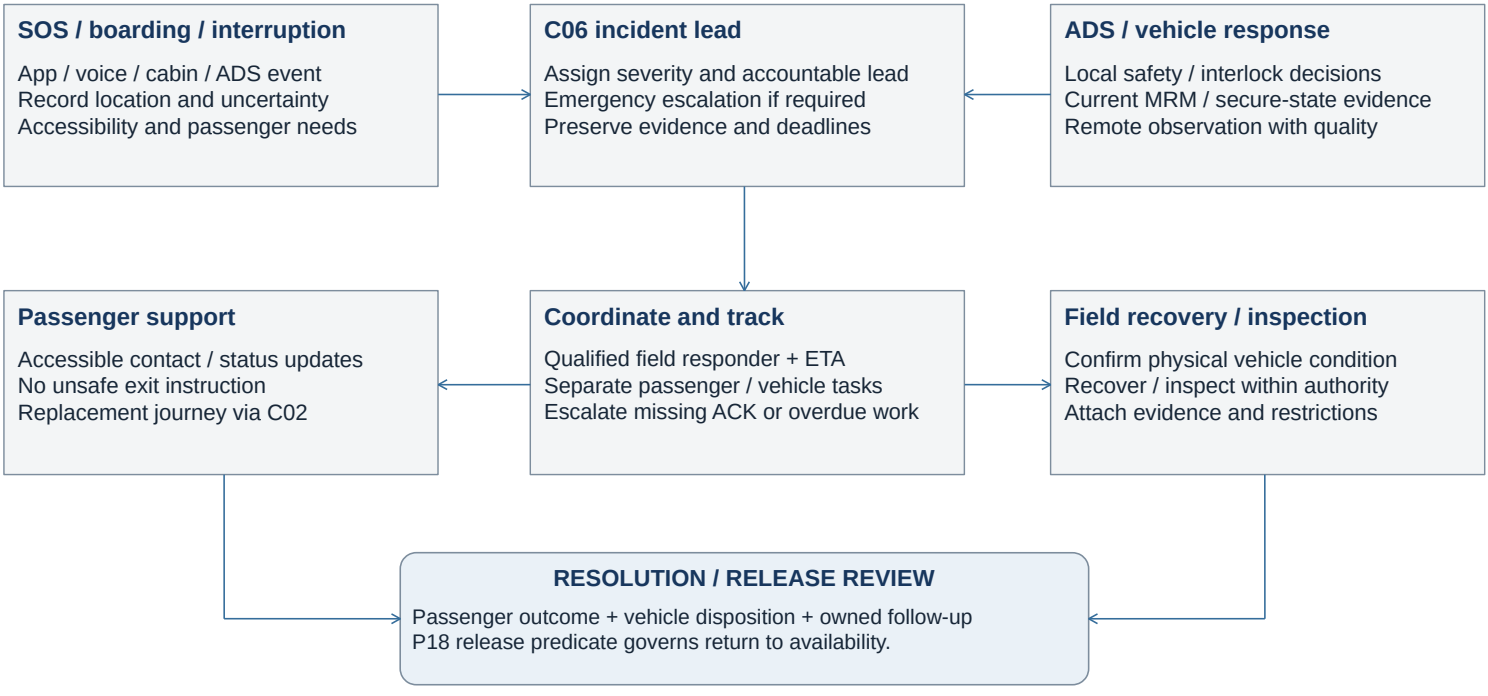
State / permitted actions

ADS mode / ODD / MRM / MRC
Motion, chassis, door, energy
Per-signal age and validity
Assess / advise / stop request
Passenger / field contact
Disabled action shows reason
Scope confirmation before submit

Action and handover record

Vehicle + mission + case revision + lease epoch + reason + advice type + expiry + expected outcome
Action-specific guards use evidence and vehicle authority. Status queries remain available with stale data.
Handover requires acceptance of pending commands, passenger status, field ETA and next deadline.
Expired / unacknowledged ownership triggers reassignment, not silent abandonment.
Relief capacity and supervisor escalation handle concurrent incidents and a failed seat.

14.2 Incident workflow with parallel vehicle and passenger outcomes



14.3 Workload and scenario acceptance

Staffing input: $A = \Sigma(\lambda_k \times E[S_k])$ in matched time units. A/m is a mean-load screen, not a service-level guarantee. Test bursts, breaks, common-cause alarms, handover and centre loss. Include field-team travel and replacement-trip capacity separately. Record detection, ACK, resolution, overdue cases and failed recovery. Restrict new commitments when response capacity is exhausted. P27 defines service-mode and return-to-service gates. Every open branch retains an owner and deadline.

Action-specific guards and time validity

RM-10 / RM-12. Permissions depend on action class. Read-only recovery and emergency escalation remain possible when telemetry is stale. Motion-enabling advice requires current contextual evidence.

15.1 Action class determines its evidence and authority requirements

Read / escalate

Authenticated status or history query
Allowed while state is UNKNOWN
No movement or allocation side effect

→

Request / advise

Current owner, fence and capability
Action-specific scene/state evidence
Stop request uses vehicle contract

→

Onboard decision

Accept/reject with explicit reason
Local ODD and safety checks prevail
No remote steering/throttle baseline

15.2 Proposed command and immutable outcome envelopes

RemoteActionRequest

tenantId, vehicleId, missionId?, commandId
caseId?, expectedCaseRev?, actionType, parameters
capabilityRev, sceneRef?, expectedVehicleStateRev?
operatorId, qualificationRef, authorityEpoch
issuedAt, expiresAt, nonce, payloadHash
reasonCode, evidenceRefs[], approvalRef?

Required fields follow a versioned action schema.
CONTEXT_ADVICE / STOP_REQUEST /
MISSION_CHANGE_REQUEST / STATUS_QUERY
Same ID with changed payload is rejected.

CommandAuditEvent

tenantId, eventId, commandId, correlationId
producerId, bootEpoch, streamId, sequence
observedAt, receivedAt, clock-bound metadata
RECEIVED / ACCEPTED / REJECTED / EXECUTING /
COMPLETED / ABORTED / EXPIRED + reason
stateRevision, evidenceRefs, policyVersion, actor

UNKNOWN belongs to the observer, not the ADS.
Keep raw event + validation result append-only.
Missing terminal evidence stays unresolved.
Expiry blocks new acceptance, not reconciliation.

15.3 Freshness uses bounded clock error and future-time rejection

$\Delta_i = t_{now} - t_{observed,i}$ $U_i = u_{source,i} + u_{observer}$
$ageUpper_i = \max(0, \Delta_i + U_i)$
$timeOK_i = boundsKnown \wedge clocksValid \wedge (\Delta_i \geq -U_i) \wedge (ageUpper_i \leq F_i)$
$valid_i = identityOK \wedge schemaOK \wedge sourceOK \wedge epoch/sequenceOK \wedge timeOK_i$
All terms use seconds. U includes synchronisation error and drift since the last trusted update.
No finite clock bound means time validity is UNKNOWN. Local timers use a monotonic clock.

Response and execution deadlines

$D_{remote} = D_{detect} + D_{queue} + D_{ack} + D_{assess} + D_{transmit} + D_{accept}$. Budget is operational. Do not use it to justify onboard safety deadlines. Expiry uses the receiver's validated time bound; clock uncertainty rejects new time-critical acceptance.

Ownership and release parameters

ADS owner: local link-loss/fallback, action capability, fence and journal persistence. Platform: F_i, clock bounds, sampling, loss, retry and retention. Operations: response class, seat/field coverage and escalation. Each value needs a unit, rationale and boundary test.

15.4 Audit scope and evidence lifecycle

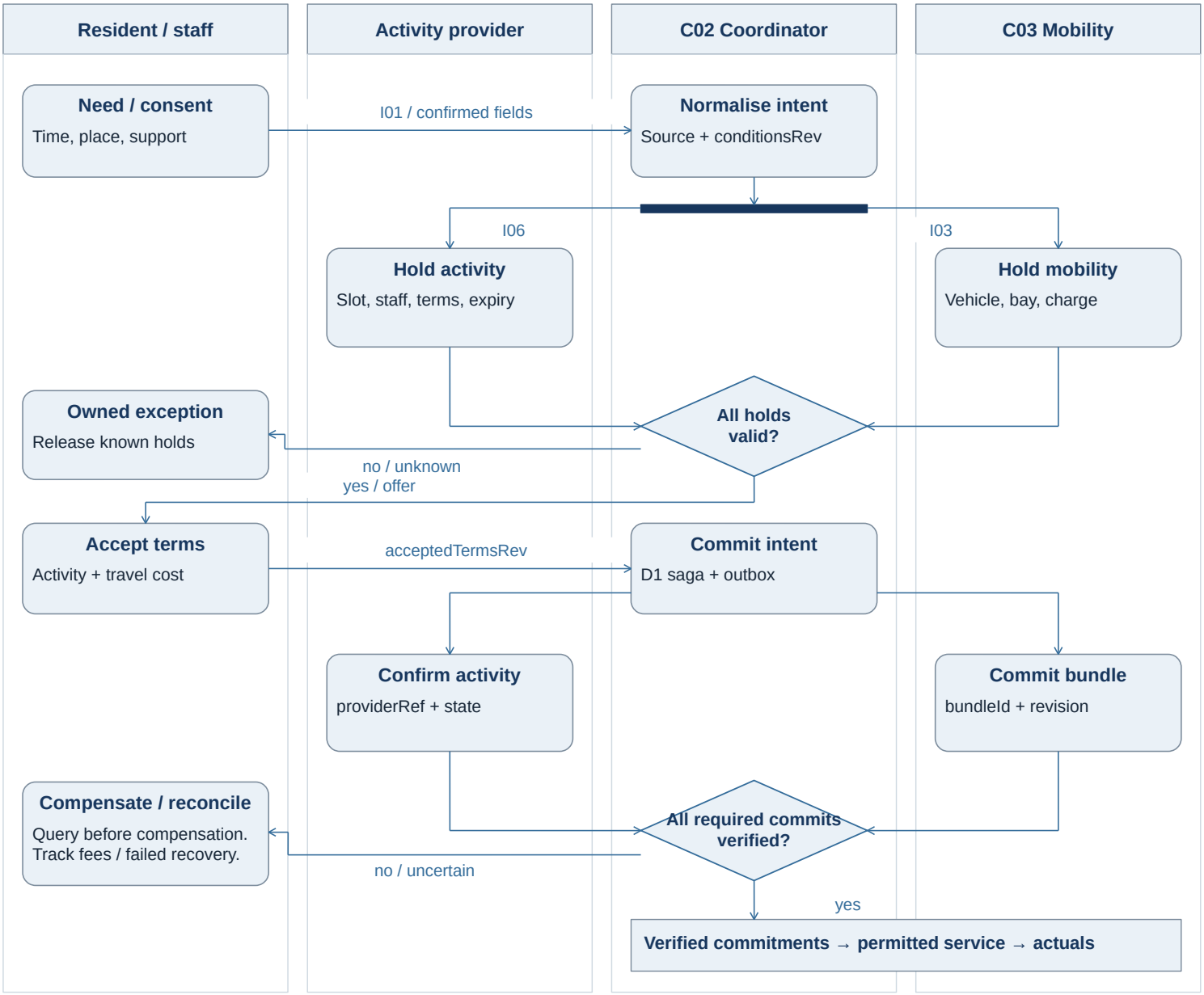
Purpose-limited capture, scoped access, encryption, retained incident evidence and read-only replay. AI may draft or summarise within task scope; it cannot independently authorise ADS movement. Test cross-tenant access, expired authority, future timestamps, unknown clock bounds and duplicate effects.

Desk-reviewed architecture proposal; implementation, performance and safety validation remain pending. Public references support terminology and patterns, not this system's certification. Sources in notes: SAE J3016 baseline, ISO 34503, NHTSA, Waymo, NPA, AWS, Microsoft, Hazelcast, PostgreSQL, RFC 5905 and calendar API documentation.

Activity saga, service workflow and AI authority

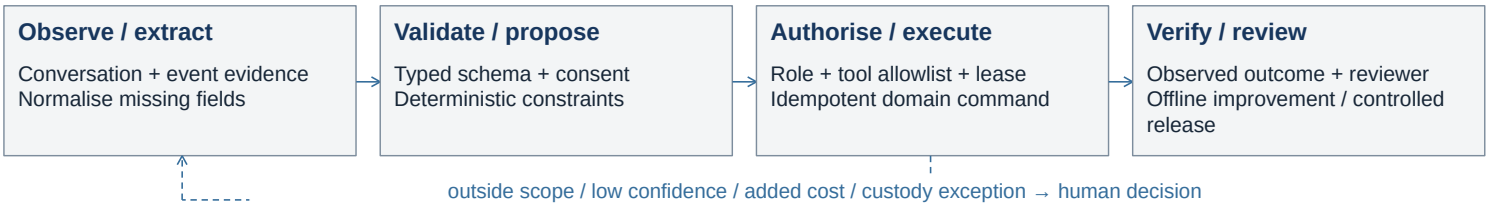
Care outings, after-school care and community events share coordination logic. Each provider retains service responsibility, staffing and handover obligations.

16.1 Swimlane activity flow with parallel holds and partial-failure compensation



Service guards: care = assistance + care-staff capacity; school care = guardian authority + verified handover; community event = venue/stop + host capacity. ETA changes re-enter the saga with a new revision. Provider and transport reservations are not one distributed transaction.

16.2 AI assistance as a constrained tool pipeline



Release stages: shadow comparison → staff-approved actions → bounded automatic tasks
Independent guards enforce consent and accepted terms. Model updates require offline evaluation and versioned release.

Calendar changes produce proposed intent, never auto-booking by default. Sync deletion/resync must not erase D1 bookings. Commercial APIs need access and hold/commit/cancel support; otherwise use an owned manual workflow.

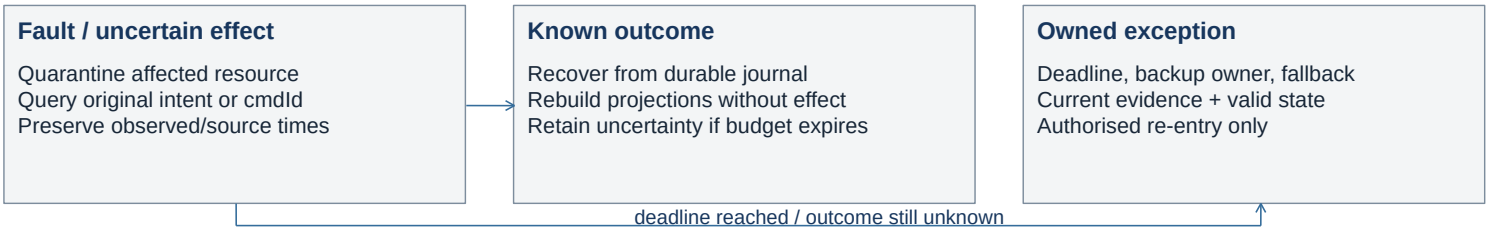
Verification coverage and release dependencies

RM-11. Remote-operation requirements join the PaaS acceptance plan. The cases below are planned tests, not passed results. Release requires reproducible evidence and operating readiness.

17.1 Fault injection and observable remote-operation invariants

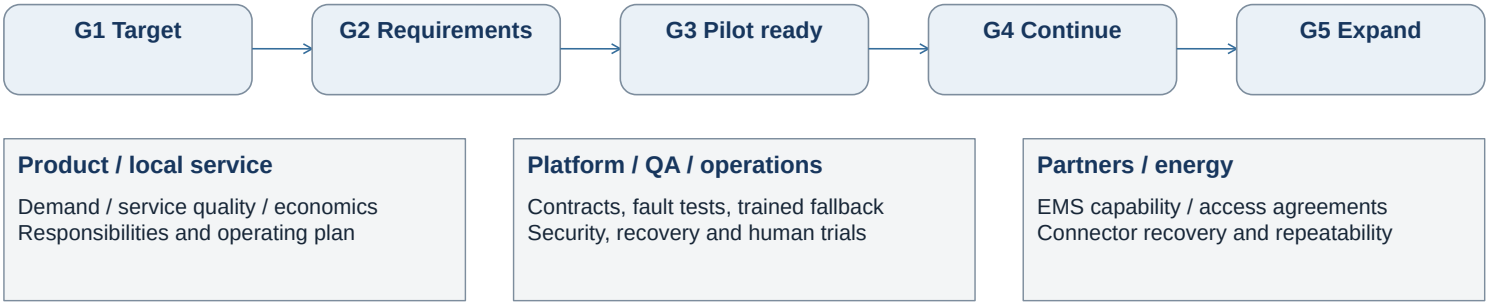
T-R01 / RM-03–04	Delay, reorder, clock jump, reboot	Stale/future data blocks motion; query stays usable.
T-R02 / RM-01,05	ODD limit / sensor degradation	Correct response, no remote extension of ODD.
T-R03 / RM-06,10	Expired / changed-scene advice	Reject with reason and renew context if required.
T-R04 / P13,RM-07	Lose ACK / replay command ID	No duplicate effect; lost result remains reconciled.
T-R05 / RM-02,07	Lose telemetry / media / voice	Independent quality flags; local policy + escalation.
T-R06 / RM-02,10	Centre partition / old operator	Fence ACK + reconcile prior accepted effects.
T-R07 / RM-05,09	SOS + MRM failure / no field team	Emergency and passenger work remain owned.
T-R08 / RM-04,08	Alarm storm / unauthorised AI	No hidden P0; deny AI motion-authorisation.

17.2 Shared PaaS invariants and fault recovery



Add roadside clock/calibration, coverage, false association, reboot and replay tests (P15–P16). P25–P28 extend acceptance to peak demand, spatial imbalance, finite hub capacity, replenishment queues and combined failures. Retain settlement regression tests.

17.3 Evidence chain, milestones and parallel workstreams



G3 evidence: scenario/ODD slice, initial state, revisions, injected fault, trace and oracle result, tested in SIL/HIL/operator/vehicle trials as applicable. G4 uses observed outcomes. G5 adds repeatable onboarding. Record owner, residual risk/restriction and signed decision at each gate.

Service capacity and geographic coverage

Operating feasibility depends on time and location. Fleet and hub totals become usable capacity only when vehicle readiness, travel time and the full service cycle align.

18.1 Demand, reachable supply and commitment

Demand by zone and time

OD requests / time window
Booking lead time and peak shape
Party size and accessibility needs
Customer cancellations recorded

→

Feasible supply

Current position and ready time
Vehicle / stop / ODD compatibility
Pickup and arrival time windows
Energy and next-task constraints

→

Service commitment

Hold the complete resource bundle
Confirm only after verified commit
Return a binding reason if infeasible
Retain unmet demand in reporting

↓

C03 evaluation at each planning interval

$t_{ready,v} + travelTime(v, origin, departure\ time) \leq latestPickup$
Next tasks include passenger travel, repositioning, turnaround and replenishment.
A reservation lead time enables physical pre-positioning only when time and energy remain feasible.

18.2 Fleet conservation and workload screen

Ready	Committed	In service	Replenishing	Unavailable
Assignable now	Held / pre-position	Pickup / passenger	Queue / energy	Fault / inspection

$$N_{total} = N_{ready} + N_{committed} + N_{service} + N_{replenishing} + N_{unavailable}$$

Necessary capacity screen

λ [requests/h] × E[T_{cycle}] [h/request] ≤ $N_{service\text{-}capable}$
T_{cycle} includes occupied and empty travel, turnaround and attributable replenishment time.
Evaluate with mutually exclusive states. Do not deduct the same downtime twice.

Spatial and peak-time check

Average capacity does not prove pickup feasibility.
Test demand bursts and directional imbalance by zone.
Hold operational reserve for faults and recovery.
Intercity transfers require explicit travel and authority.

18.3 Hub function, allocation and service outcome

Hub resources

Parking bays and pickup space
Connectors / dispensers / site power
Cleaning and inspection positions
Qualified staff and opening hours

→

Finite allocation

Use time intervals per resource
Sum pooled demand against limits
Reserve only accessible capacity
Propagate queues and outage derating

→

Observable outcome

Dispatch / pickup / completion
Unserved and delayed requests
Queue and recovery workload
Energy and cost per completed trip

Planning record: zone / time interval, OD input, initial vehicle positions, hub resources, availability policy and service windows. Compare feasible alternatives under the same demand trace. A hub count alone does not specify throughput or coverage.

Replenishment and departure readiness

The operating plan reserves replenishment capacity within a vehicle's duty cycle. Daily energy sufficiency and on-time departure are separate acceptance conditions.

19.1 Vehicle cycle with finite replenishment resources

Arrive
Observed energy
Arrival time

→

Queue
Bay / connector
Staff eligibility

→

Replenish
Power or fuel
Actual delivery

→

Turnaround
Clean / inspect
Required evidence

→

Release
Energy + tasks
Next departure

Departure constraint

$t_{arrival} + t_{queue} + t_{replenish} + t_{required_work} \le t_{departure}$
Only explicitly approved concurrent tasks may overlap. Travel to a different site consumes time and energy.
A late or uncertain delivery triggers re-planning before the customer promise becomes infeasible.

19.2 Electrical and fuel branches use distinct units

Electrical charging

$E_{depart} = E_{arrive} + \eta_{ch} \int P_{AC}(t)\,dt - E_{aux}$
$E_{next_trip} + E_{reserve} \le E_{depart} \le E_{usable,max}$
P_{AC} in kW, time in hours, energy in kWh.
Use charger / battery limits and charging efficiency.
Energy plan ACK does not prove delivered energy.

Fuel / hydrogen replenishment

$m_{depart} = m_{arrive} + m_{delivered} - m_{consumed}$
$m_{next_trip} + m_{reserve} \le m_{depart} \le m_{usable,max}$
Mass in kg, flow in kg/min, duration in minutes.
Bound usable stock, dispenser flow and turnaround.
Fuel-to-energy conversion needs an explicit model.

SITE LIMITS AT EVERY TIME INTERVAL

$\Sigma P_{charge,i}(t) \le P_{site,available}(t)$ and occupied connectors ≤ serviceable connectors
$P_{site,available}$ reflects building load, generation, storage and connection limits at the same boundary.
Fuel stock remains non-negative after deliveries and withdrawals. One hub can contain several resources.

19.3 Shortage propagation and recovery

Detect
Power derating / site outage
Low stock / supply delay
Queue growth / delayed arrival
Stale capability or meter data

→

Re-plan
Recalculate departure readiness
Use feasible alternative site / vehicle
Hold affected assignments
Revise ETA and notify service owner

→

Reconcile and release
Observe actual delivered quantity
Complete inspection and required work
Confirm current resource revisions
Release only verified capacity

Acceptance evidence: arrival / departure energy, measured delivery, queue duration, site-limit trace and missed departures. Test concurrent arrivals, one resource unavailable, reduced power and delayed fuel delivery. Report reserve violations and unmet trips even when aggregate daily energy balances.

Degraded operation and recovery capacity

Operational continuity requires bounded service reductions, passenger support and a verified return to service. Recovery uses the same finite fleet, site and staffing resources.

20.1 Failure propagation across the service system

Demand pressure

Peak arrivals / directional imbalance
Longer trips / slow turnaround

Supply loss

Vehicle fault / site outage / energy shortfall
Communication loss / staff absence

↓

Impact assessment

Identify affected bookings and intervals. Recompute reachable vehicles, usable hubs and qualified response capacity.
Preserve already accepted commitments and unknown physical effects. Record unmet demand and passenger exposure.

↓

NORMAL

Service promise feasible
Capacity and evidence current

→

CONSTRAINED

Restrict new offers by zone / time
Protect accepted journeys

→

SUSPENDED

Stop affected new commitments
Maintain support and local fallback

Transitions use configured triggers, persistence and owner approval. Recovery thresholds include hysteresis to avoid repeated entry / exit. Local ADS safety decisions remain independent of this service-mode model.

20.2 Parallel passenger, asset and capacity recovery

Passenger outcome

Contact through accessible channel
Explain revised ETA / alternatives
Arrange feasible replacement journey
Track handover and resolution
Own compensation / refund decision

Vehicle / site outcome

Retain fault and command history
Assign qualified field responder
Observe secure state and clearance
Inspect and record restrictions
Reconcile unresolved physical effects

Resource outcome

Reserve replacement and field capacity
Include responder travel and queue
Recalculate affected bookings
Track backlog age and deadlines
Escalate resource exhaustion

20.3 Return-to-service gate and staffing condition

Return-to-service gate

Cause addressed + fresh evidence + authorised release + feasible next mission + passenger outcome resolved.
Connectivity restoration alone cannot release capacity.
Observe sustained recovery before widening offers.

Response capacity

$A = \Sigma \lambda_k E[S_k]$ in matched time units.
$A <$ qualified staffed seats is only a mean-load screen.
Test bursts, common-cause alarms, breaks and travel.
Every open case needs an owner, deadline and backup.
Keep field response separate from remote assistance.

Record trigger, affected scope, admission restriction, accountable owner, next review and recovery evidence for each degraded period.

Operating feasibility and verification evidence

A configuration qualifies only within its tested demand, geography and operating conditions. Service quality, resource limits, recovery and economics require separate evidence.

21.1 Measurement chain and explicit denominators

Requests
Unique service intents

→

Dispatch
Verified assignment

→

Pickup
Physical arrival

→

Completion
Verified trip outcome

Service quality

Dispatch success = assigned / eligible dispatch requests.
Keep capacity and admission rejects in the denominator.
Demand fulfillment = completed / eligible requests.
Report customer cancellations separately by reason.

Pickup and utilization

Wait starts at request time for immediate demand, or agreed pickup time for a reservation. Report p50 / p95 with the observed-arrival count and missing outcomes.
Show on-time pickups / pickups due, including failures.
Separate available hours from scheduled fleet hours.

Count each request once at the reporting cut-off: completed, cancelled, rejected, expired, unresolved or still active. Preserve event history separately. Request cohorts and deadlines determine metric eligibility. Show exclusions, censoring and missing evidence.

21.2 Reproducible scenario design and pass conditions

Freeze inputs

OD / hourly arrivals and trip lengths
Booking share and lead-time profile
Initial fleet positions / availability
Hub, energy and response limits
Geography and parameter revisions

→

Exercise conditions

Normal demand and forecast error
Peak and directional concentration
Vehicle / hub / energy capacity loss
Link outage and support overload
Combined faults and recovery ramp

→

Evaluate each case

Service targets by zone and time
Resource and energy conservation
Queue age and passenger recovery
Missing / unresolved outcomes
Binding constraint and owner

PASS RECORD

State target, denominator, time window and acceptance rule before running. Use repeatable demand traces or recorded seeds, then independent trials. Preserve every run and failure. Finite synthetic tests establish a tested envelope only. Pilot evidence determines field readiness.

21.3 Capacity decision and business consequence

Feasible alternatives

Compare fleet, hub placement and service policy only after all required operating conditions pass.
Rank cost within the feasible tested set. If none qualifies, report the binding constraint and re-plan.

Economic reconciliation

Use completed trips and measured resource use.
Include empty travel, energy, support and recovery.
Separate CAPEX, annual operating cost and cash flow.
Avoid double-counting integrated hub infrastructure.

Release evidence: input revision, scenario / seed, execution trace, metric definitions, pass / fail by condition, residual restriction and accountable decision. Operational feasibility does not by itself establish project profitability or safety certification.